supplemental material for

Transition from antiferromagnets to altermagnets: symmetry breaking theory

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I. Symmetry relationship between collinear antiferromagnets and altermagnets

A. Groups and subgroups of spin point groups in collinear magnets

A spin point group[1], denoted by $\mathcal{R}_s$, consists of elements of the form $[R_i\|R_j]$, where the left-hand component (R_i) belongs to the group $\mathcal{B}$ and the right-hand component (R_j) belongs to the group $\mathcal{G}$. Both $\mathcal{B}$ and $\mathcal{G}$ are members of one of the 32 crystallographic point group types, and they are, respectively, represent the operations in spin space and real atomic space. Since we are focused on non-relativistic collinear magnets, the possible point groups for the group $\mathcal{B}$ are 1, $\bar{1}$, 2, and m. When $\mathcal{B} = 1$, the subgroup $\mathcal{N}$ of the point group $\mathcal{G}$ coincides with $\mathcal{G}$ itself, resulting in the spin point group $\mathcal{R}_s = [E\|\mathcal{N}]$. Consequently, there are 32 such spin point groups in total, corresponding to collinear ferromagnets. When $\mathcal{B}$ is $\bar{1}$, 2, or m, the spin point group takes the form $\mathcal{R}_s = [E\|\mathcal{N}]+[B_2\|R_2][E\|\mathcal{N}]$. Here, B_2 can represent inversion symmetry P, a 180° rotation symmetry C_2 perpendicular to the spin arrangement, or a mirror symmetry M parallel to the spin arrangement. Since these symmetries are equivalent in collinear magnets, we conventionally use C_2 symmetry to characterize the spin point groups. In this case, the group $\mathcal{B}$ is 2, and the spin point group is denoted as

$$\mathcal{R}_s = [E\|\mathcal{N}] + [C_2\|R_2][E\|\mathcal{N}] = [E\|\mathcal{N}] + [C_2\|\mathcal{G}-\mathcal{N}], \tag{1}$$

where the normal subgroup $\mathcal{N}$ is a halving subgroup, and $\mathcal{G}-\mathcal{N}$ forms the coset $\mathcal{AN}$. As shown in Tab. S1, there are 58 such spin point groups. When the symmetry $[C_2\|P]$, which preserves the degeneracy between spin-up and spin-down states, is incorporated into the spin point group $\mathcal{R}_s$, it describes PT-symmetric antiferromagnets. These are represented by the 21 spin point groups marked in light red in Tab. S1. In contrast, the remaining 37 spin point groups, highlighted in green in Tab. S1, correspond to altermagnets. When the spin point groups describing antiferromagnets and altermagnets satisfy a group-subgroup symmetry relationship, the PT antiferromagnetic phase can be transformed into the altermagnetic phase through symmetry breaking. As demonstrated in Tab. S2 and Tab. S3, there are 82 symmetry-breaking paths that enable the transition from the antiferromagnetic phase to the altermagnetic phase.

In addition to PT-symmetric antiferromagnetism, $\boldsymbol{t}T$-symmetric antiferromagnetism constitutes another major category of conventional collinear antiferromagnets. However, since $\boldsymbol{t}T$ symmetry inherently involves a translational component $\boldsymbol{t}$, it cannot be fully captured by the spin point group alone. To rigorously describe its symmetry, we introduce two distinct translations: $\boldsymbol{t}_1$, acting within the same spin sublattice, and $\boldsymbol{t}_2$, connecting opposite spin sublattices. These can be represented in spin group notation as $[E\|(E|\boldsymbol{t}_1)]$ and $[C_2\|(E|\boldsymbol{t}_2)]$, respectively. Together, these elements define the modified spin-only group $r'_s = \{[E\|(E|\boldsymbol{t}_1)], [C_2\|(E|\boldsymbol{t}_2)]\}$. Next, we consider the inclusion of a non-trivial spin group $\mathcal{R}_s$. If $\mathcal{R}_s = [E\|\mathcal{N}]$, the first-class spin space group of the $\boldsymbol{t}T$ antiferromagnet can be constructed as

$$\begin{aligned}\mathcal{R}^{I}_{SG} &= r'_s \times \mathcal{R}_s \\ &= \{[E\|(E|\boldsymbol{t}_1)], [C_2\|(E|\boldsymbol{t}_2)]\} \times \{[E\|\mathcal{N}]\} \\ &= [E\|(E|\boldsymbol{t}_1)][E\|\mathcal{N}] + [C_2\|(E|\boldsymbol{t}_2)][E\|\mathcal{N}] \\ &= [E\|(\mathcal{N}|\boldsymbol{t}_1)] + [C_2\|(\mathcal{N}|\boldsymbol{t}_2)].\end{aligned} \tag{2}$$

The presence of $[C_2\|(E|\boldsymbol{t}_2)]$, equivalent to $\boldsymbol{t}T$ symmetry, enforces spin degeneracy across the entire Brillouin zone. Upon breaking the translation $\boldsymbol{t}_2$, the system transitions to the spin group $\mathcal{R}^{I}_{SG'} = [E\|(\mathcal{N}|\boldsymbol{t}_1)]$, corresponding to the ferromagnetic state. Thus, in this case, the $\boldsymbol{t}T$ antiferromagnet cannot transition into an altermagnet.

However, when the spin group is $\mathcal{R}_s = [E\|\mathcal{N}] + [C_2\|(\mathcal{G}-\mathcal{N})]$, the second-class spin space group can be defined as

$$\begin{aligned}\mathcal{R}^{II}_{SG} &= r'_s \times \mathcal{R}_s \\ &= \{[E\|(E|\boldsymbol{t}_1)], [C_2\|(E|\boldsymbol{t}_2)]\} \times \{[E\|\mathcal{N}], [C_2\|\mathcal{G}-\mathcal{N}]\} \\ &= [E\|(E|\boldsymbol{t}_1)][E\|\mathcal{N}] + [E\|(E|\boldsymbol{t}_1)][C_2\|\mathcal{G}-\mathcal{N}] + [C_2\|(E|\boldsymbol{t}_2)][E\|\mathcal{N}] + [C_2\|(E|\boldsymbol{t}_2)][C_2\|\mathcal{G}-\mathcal{N}] \\ &= [E\|(\mathcal{N}|\boldsymbol{t}_1)] + [C_2\|(\mathcal{G}-\mathcal{N}|\boldsymbol{t}_1)] + [C_2\|(\mathcal{N}|\boldsymbol{t}_2)] + [E\|(\mathcal{G}-\mathcal{N}|\boldsymbol{t}_2)].\end{aligned} \tag{3}$$

Upon breaking the $\boldsymbol{t}_2$ symmetry, this group transforms into $\mathcal{R}^{II}_{SG'} = [E\|(\mathcal{N}|\boldsymbol{t}_1)] + [C_2\|(\mathcal{G}-\mathcal{N}|\boldsymbol{t}_1)]$, which corresponds to either PT-symmetric antiferromagnets or altermagnets, depending on the remaining symmetries. In the case of PT-symmetric antiferromagnetism, further breaking of the PT symmetry is required to enable the transition from $\boldsymbol{t}T$ antiferromagnetism to altermagnetism. Based on the above discussion, Tab. S4 summarizes all collinear spin space groups capable of realizing the transition to altermagnetism via $\boldsymbol{t}T$ symmetry breaking.

B. The details about all pathways for magnetic transition in PT-symmetric antiferromagnets

The symmetries of collinear magnetic materials without spin-orbit coupling can be effectively characterized using spin point groups. Among these, we identify 21 non-trivial spin point groups derived from 11 distinct crystallographic point groups, where the inversion symmetry P lies in the coset $\mathcal{G}-\mathcal{N}$ (highlighted in light red in Tab. S1). These spin groups correspond to antiferromagnets with preserved PT symmetry. To systematically explore all possible pathways for transitions from antiferromagnetism to altermagnetism, we categorize the 21 spin point groups into 11 classes based on their parent crystallographic point groups. For each class, we examine the symmetry-breaking processes within the corresponding non-trivial spin groups and derive their subgroups. Through this systematic analysis, we identify 82 potential transition pathways leading from antiferromagnetic to altermagnetic phases. A summary of these results is provided in Tabs. S2 and S3. In the sections that follow, we present a detailed discussion of the symmetry evolution for each of the 21 non-trivial spin point groups.

1. *The crystallographic point group* $\bar{1}$

The parent crystallographic point group $\bar{1}$ generates the spin point group ${}^{2}\bar{1}$, which represents an antiferromagnetic state. In this case, the coset of ${}^{2}\bar{1}$ includes only the inversion symmetry P, denoted as $[C_2\|P]$. Once this symmetry is broken, there remains no symmetry connecting the opposite spin sublattices, leading the system to transition directly into the ferromagnetic phase. As a result, the antiferromagnetic phase characterized by ${}^{2}\bar{1}$ does not permit a symmetry-breaking pathway toward altermagnetism, as illustrated in Fig. S1 and summarized in Tab. S2.

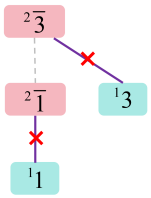


Fig. S1: The spin group evolution of the crystallographic point group $\bar{1}$ and $\bar{3}$.

2. *The crystallographic point group* $2/m$

The parent crystallographic point group $2/m$ gives rise to three spin point groups: ${}^{2}2/{}^{2}m$, ${}^{1}2/{}^{2}m$, and ${}^{2}2/{}^{1}m$. Among these, ${}^{2}2/{}^{2}m$ corresponds to an altermagnetic phase, whereas ${}^{1}2/{}^{2}m$ and ${}^{2}2/{}^{1}m$ include inversion symmetry P in their cosets and thus describe conventional antiferromagnetic states. For the spin group ${}^{1}2/{}^{2}m$, breaking the symmetries $[C_2\|P]$ and $[E\|C_2]$ results in the altermagnetic spin group ${}^{2}m$. Likewise, breaking $[C_2\|P]$ and $[E\|M]$ in ${}^{2}2/{}^{1}m$ leads to the altermagnetic spin group ${}^{2}2$. Accordingly, the $2/m$ point group supports two distinct symmetry-breaking pathways from antiferromagnetism to altermagnetism, as illustrated in Fig. S2 and summarized in Tab. S2.

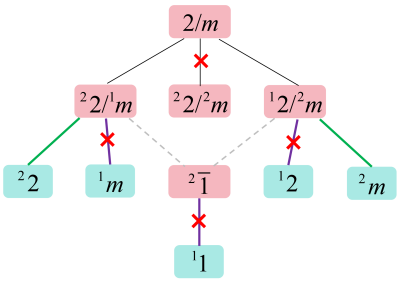


Fig. S2: The spin group evolution of the crystallographic point group $2/m$.

3. The crystallographic point group mmm

The parent crystallographic point group mmm gives rise to three spin point groups: ${}^{2}m^{2}m^{1}m$, ${}^{1}m^{1}m^{2}m$, and ${}^{2}m^{2}m^{2}m$. Among these, ${}^{2}m^{2}m^{1}m$ corresponds to altermagnetic phases, while ${}^{1}m^{1}m^{2}m$ and ${}^{2}m^{2}m^{2}m$ include inversion symmetry P in their cosets, thus representing conventional antiferromagnetic states. For ${}^{1}m^{1}m^{2}m$, breaking the symmetries $[C_2\|P]$, $[C_2\|C_{2y}]$, $[E\|M_y]$, and $[E\|C_{2z}]$ (or alternatively, $[C_2\|P]$, $[C_2\|M_z]$, $[E\|M_x]$, and $[E\|M_y]$) leads to the altermagnetic spin group ${}^{2}m^{1}m^{2}2$ (or ${}^{2}2^{2}2^{1}2$). This spin group may also undergo a two-step reduction: first breaking selected rotational or mirror symmetries while retaining inversion P to reach intermediate subgroups such as ${}^{1}2/{}^{2}m$ or ${}^{2}2/{}^{1}m$, and then breaking P to arrive at the altermagnetic phase, following the pathway similar to that of the $2/m$ point group. For ${}^{2}m^{2}m^{2}m$, breaking $[C_2\|P]$, $[C_2\|M_z]$, $[E\|C_{2x}]$, and $[E\|C_{2y}]$ yields the altermagnetic spin group ${}^{2}m^{2}m^{1}2$. Alternatively, breaking $[C_2\|M_x]$, $[C_2\|M_y]$, $[E\|C_{2x}]$, and $[E\|C_{2y}]$ results in the intermediate spin group ${}^{1}2/{}^{2}m$, which can subsequently break inversion P to form the altermagnetic spin group ${}^{2}m$. In total, the mmm point group enables six distinct symmetry-breaking pathways from antiferromagnetic to altermagnetic states, as illustrated in Fig. S3 and summarized in Tab. S2.

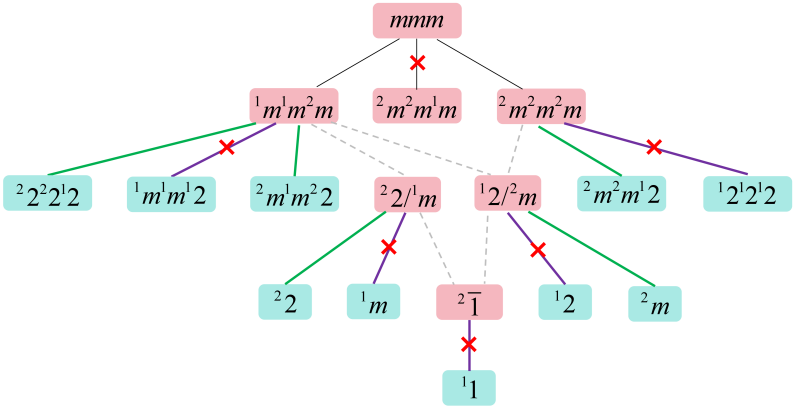


Fig. S3: The spin group evolution of the crystallographic point group mmm.

4. The crystallographic point group $4/m$

The parent crystallographic point group $4/m$ gives rise to three spin point groups: ${}^{2}4/{}^{1}m$, ${}^{2}4/{}^{2}m$, and ${}^{1}4/{}^{2}m$. Among them, ${}^{2}4/{}^{1}m$ corresponds to altermagnetic phases, while the other two, ${}^{2}4/{}^{2}m$ and ${}^{1}4/{}^{2}m$, contain inversion symmetry P in their cosets and thus represent conventional antiferromagnetic states. For the spin group ${}^{2}4/{}^{2}m$, breaking the symmetries $[C_2\|P]$ and $[E\|\bar{C}_{4z}]$ leads to the altermagnetic spin group ${}^{2}4$. Similarly, breaking $[C_2\|P]$ and $[E\|C_{4z}]$ in ${}^{1}4/{}^{2}m$ results in the altermagnetic spin group ${}^{2}\bar{4}$. In both cases, an alternative two-step route exists: first, breaking the fourfold (or roto-inversion) symmetry while preserving inversion P yields the intermediate antiferromagnetic group ${}^{1}2/{}^{2}m$, which can then undergo inversion symmetry breaking to reach the altermagnetic phase, resembling the symmetry-breaking process observed in the $2/m$ point group. Altogether, the $4/m$ point group supports four distinct symmetry-breaking pathways from antiferromagnetism to altermagnetism, as shown in Fig. S4 and summarized in Tab. S2.

5. The crystallographic point group $4/mmm$

The parent crystallographic point group $4/mmm$ gives rise to five spin point groups: ${}^{2}4/{}^{1}m^{1}m^{2}m$, ${}^{1}4/{}^{1}m^{2}m^{2}m$, ${}^{2}4/{}^{2}m^{2}m^{1}m$, ${}^{1}4/{}^{2}m^{1}m^{1}m$, and ${}^{1}4/{}^{2}m^{2}m^{2}m$. Among these, ${}^{2}4/{}^{1}m^{1}m^{2}m$ and ${}^{1}4/{}^{1}m^{2}m^{2}m$ correspond to altermagnetic phases, while the remaining three, ${}^{2}4/{}^{2}m^{2}m^{1}m$, ${}^{1}4/{}^{2}m^{1}m^{1}m$, and ${}^{1}4/{}^{2}m^{2}m^{2}m$, retain inversion symmetry P in their cosets and thus represent conventional antiferromagnets.

These antiferromagnetic spin groups can transition into altermagnetic phases via specific symmetry breakings. For instance, breaking $[C_2\|P]$, $[C_2\|C_{4z}^{+}]$, $[E\|C_{2x}]$, and $[E\|M_{xy}]$ in ${}^{2}4/{}^{2}m^{2}m^{1}m$ yields the altermagnetic group ${}^{1}\bar{4}^{2}2^{2}m$.

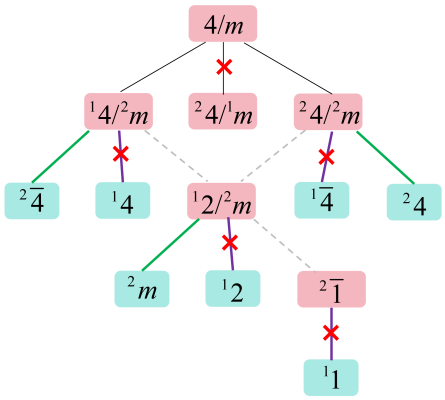


Fig. S4: The spin group evolution of the crystallographic point group $4/m$.

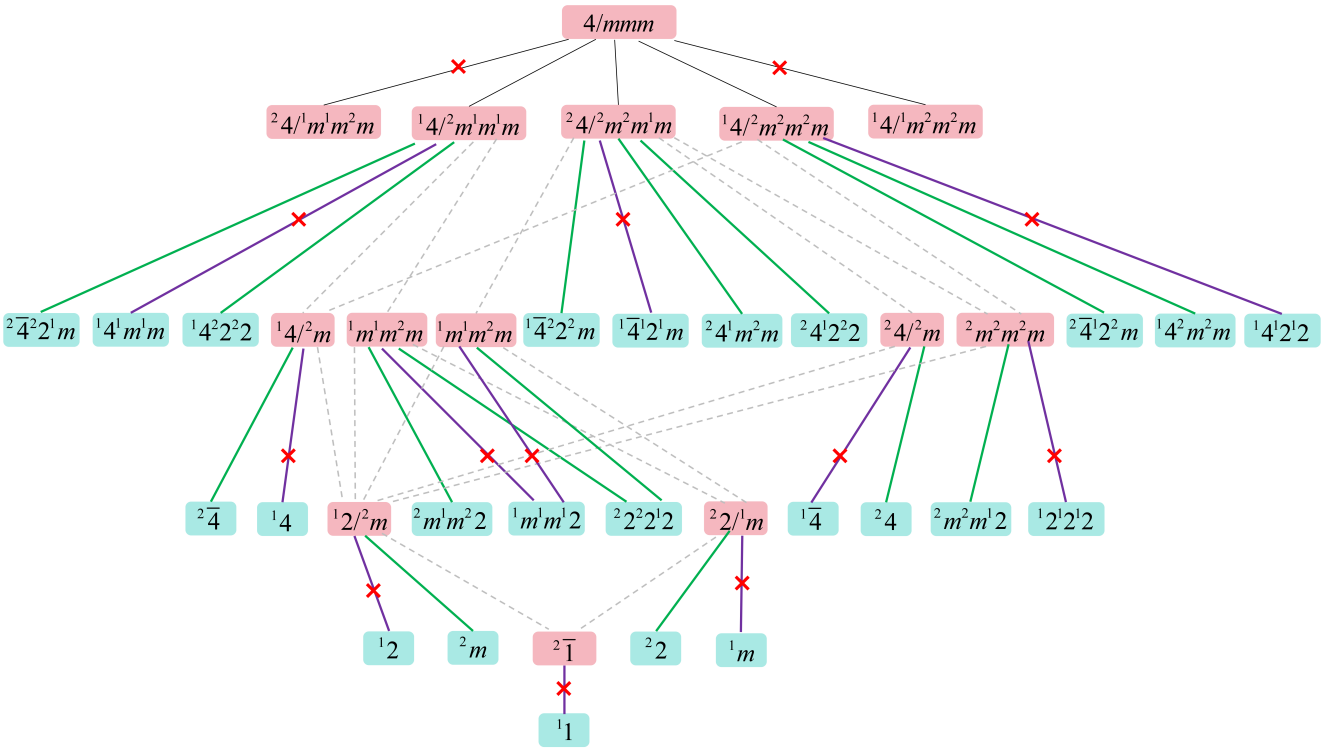


Fig. S5: The spin group evolution of the crystallographic point group $4/mmm$.

Alternatively, breaking $[C_2\|P]$, $[C_2\|C_{2xy}]$, $[E\|C_{2x}]$, and $[E\|\bar{C}^{+}_{4z}]$ leads to $^{2}4^{1}m^{2}m$, while the combination $[C_2\|P]$, $[C_2\|M_x]$, $[E\|M_{xy}]$, and $[E\|\bar{C}^{+}_{4z}]$ results in $^{2}4^{1}2^{2}2$. This group can also evolve via intermediate subgroups such as $^{1}4/^{2}m$, $^{2}4/^{2}m$, $^{1}m^{1}m^{2}m$, $^{2}m^{2}m^{2}m$, $^{1}2/^{2}m$, and $^{2}2/^{1}m$, all of which preserve P and can transition into altermagnetic states upon further breaking P. In total, $^{2}4/^{2}m^{2}m^{1}m$ offers eight symmetry-breaking routes.

Similarly, $^{1}4/^{2}m^{1}m^{1}m$ can transform into the altermagnetic group $^{2}\bar{4}^{2}2^{1}m$ by breaking $[C_2\|P]$, $[C_2\|C_{2xy}]$, $[E\|M_x]$, and $[E\|C^{+}_{4z}]$, or yield $^{1}4^{2}2^{2}2$ by breaking $[C_2\|P]$, $[C_2\|\bar{C}^{+}_{4z}]$, $[E\|M_x]$, and $[E\|M_{xy}]$. Like the previous case, this group may follow a two-step path through intermediate subgroups ($^{1}4/^{2}m$, $^{1}m^{1}m^{2}m$, $^{1}2/^{2}m$, $^{2}2/^{1}m$), then break inversion symmetry to reach altermagnetic phases. In total, seven such pathways exist. For $^{1}4/^{2}m^{2}m^{2}m$, breaking $[C_2\|P]$, $[C_2\|M_x]$, $[E\|C_{2xy}]$, and $[E\|C^{+}_{4z}]$ yields the altermagnetic group $^{2}\bar{4}^{1}2^{2}m$, while breaking $[C_2\|P]$, $[C_2\|\bar{C}^{+}_{4z}]$, $[E\|C_{2x}]$, and $[E\|C_{2xy}]$ leads to $^{1}4^{2}m^{2}m$. This group can also reach intermediate subgroups like $^{1}4/^{2}m$, $^{2}m^{2}m^{2}m$, or $^{1}2/^{2}m$, and then transition to altermagnetism by breaking P, resulting in five total paths.

In summary, the three antiferromagnetic spin point groups, $^{2}4/^{2}m^{2}m^{1}m$, $^{1}4/^{2}m^{1}m^{1}m$, and $^{1}4/^{2}m^{2}m^{2}m$, collec-

tively support 20 distinct symmetry-breaking pathways toward altermagnetic states, as illustrated in Fig. S5 and summarized in Tab. S2.

6. *The crystallographic point group* $\bar{3}$

The parent crystallographic point group $\bar{3}$ gives rise to the spin point group ${}^{2}\bar{3}$, which corresponds to the antiferromagnetic phase. In ${}^{2}\bar{3}$, the coset representatives include both inversion symmetry P and threefold roto-inversion $\bar{C}_3$, denoted as $[C_2\|P]$ and $[C_2\|\bar{C}_3]$. Breaking the inversion symmetry P leads to the subgroup ${}^{1}3$, associated with ferromagnetism. In contrast, breaking the $\bar{C}_3$ symmetry results in the subgroup ${}^{2}\bar{1}$, which, as discussed previously for the $\bar{1}$ point group, offers no feasible pathway from antiferromagnetism to altermagnetism. Consequently, the antiferromagnetic phase described by ${}^{2}\bar{3}$ does not support any symmetry-breaking transition into the altermagnetic phase, as summarized in Fig. S1 and Tab. S2.

7. *The crystallographic point group* $\bar{3}m$

The parent crystallographic point group $\bar{3}m$ generates three spin point groups: ${}^{1}\bar{3}^{2}m$, ${}^{2}\bar{3}^{1}m$, and ${}^{2}\bar{3}^{2}m$. Among them, ${}^{1}\bar{3}^{2}m$ corresponds to altermagnetic phases, whereas ${}^{2}\bar{3}^{1}m$ and ${}^{2}\bar{3}^{2}m$ retain inversion symmetry P in their cosets and thus represent conventional antiferromagnetic phases. For the spin group ${}^{2}\bar{3}^{1}m$, breaking the symmetries $[C_2\|P]$, $[E\|\bar{C}_3]$, together with the three vertical mirror planes, leads to the altermagnetic spin group ${}^{1}3^{2}2$. Similarly, for ${}^{2}\bar{3}^{2}m$, breaking $[C_2\|P]$, $[E\|\bar{C}_3]$, along with the three in-plane twofold rotations, yields the altermagnetic spin group ${}^{1}3^{2}m$. Furthermore, both ${}^{2}\bar{3}^{1}m$ and ${}^{2}\bar{3}^{2}m$ can first break the C_3 rotational symmetry while preserving inversion, resulting in the spin groups ${}^{2}2/{}^{1}m$ and ${}^{1}2/{}^{2}m$, respectively. Subsequent breaking of inversion symmetry P within these subgroups facilitates the transition to the altermagnetic phase, resembling the symmetry-breaking mechanism observed in the $2/m$ point group. In total, the point group $\bar{3}m$ supports four distinct symmetry-breaking pathways enabling the transition from antiferromagnetism to altermagnetism, as summarized in Fig. S6 and Tab. S2.

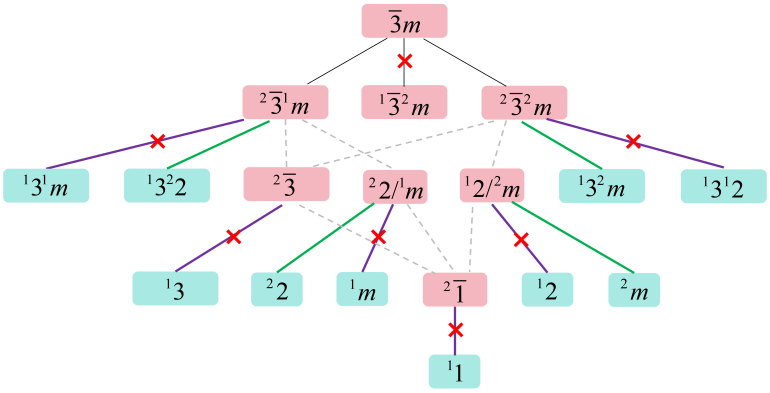


Fig. S6: The spin group evolution of the crystallographic point group $\bar{3}m$.

8. *The crystallographic point group* $6/m$

The parent crystallographic point group $6/m$ generates three spin point groups: ${}^{2}6/{}^{2}m$, ${}^{2}6/{}^{1}m$, and ${}^{1}6/{}^{2}m$. Among these, ${}^{2}6/{}^{2}m$ corresponds to altermagnetic phases, while ${}^{2}6/{}^{1}m$ and ${}^{1}6/{}^{2}m$ possess inversion symmetry P within their cosets and thus describe conventional antiferromagnetic states. For the spin group ${}^{2}6/{}^{1}m$, breaking the symmetries $[C_2\|P]$, $[C_2\|\bar{C}_3]$, and $[E\|\bar{C}_6]$ results in the altermagnetic spin group ${}^{2}6$. Similarly, for ${}^{1}6/{}^{2}m$, breaking $[C_2\|P]$, $[C_2\|\bar{C}_6]$, and $[E\|C_6]$ leads to the altermagnetic spin group ${}^{2}\bar{6}$. Furthermore, both ${}^{2}6/{}^{1}m$ and ${}^{1}6/{}^{2}m$ allow for a two-step transition route: first reducing the rotational symmetry by breaking C_3 and C_6 while preserving inversion,

thereby yielding intermediate spin groups ${}^{2}2/{}^{1}m$ and ${}^{1}2/{}^{2}m$, respectively; then, breaking inversion symmetry P to complete the transition into altermagnetic states, following the same mechanism discussed for the $2/m$ point group. Consequently, the $6/m$ point group supports four distinct symmetry-breaking pathways bridging antiferromagnetic and altermagnetic phases, as summarized in Fig. S7 and Tab. S2.

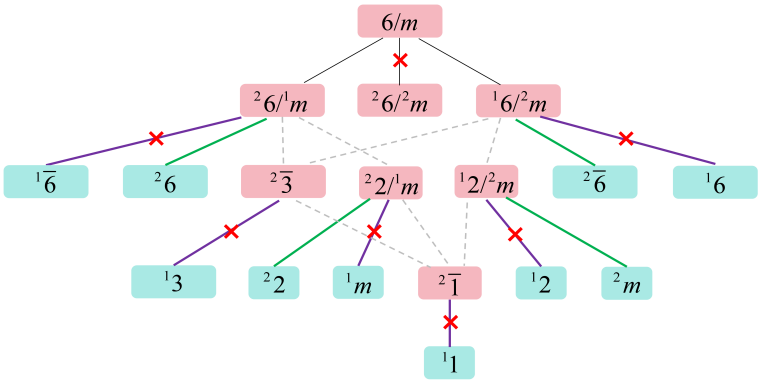


Fig. S7: The spin group evolution of the crystallographic point group $6/m$.

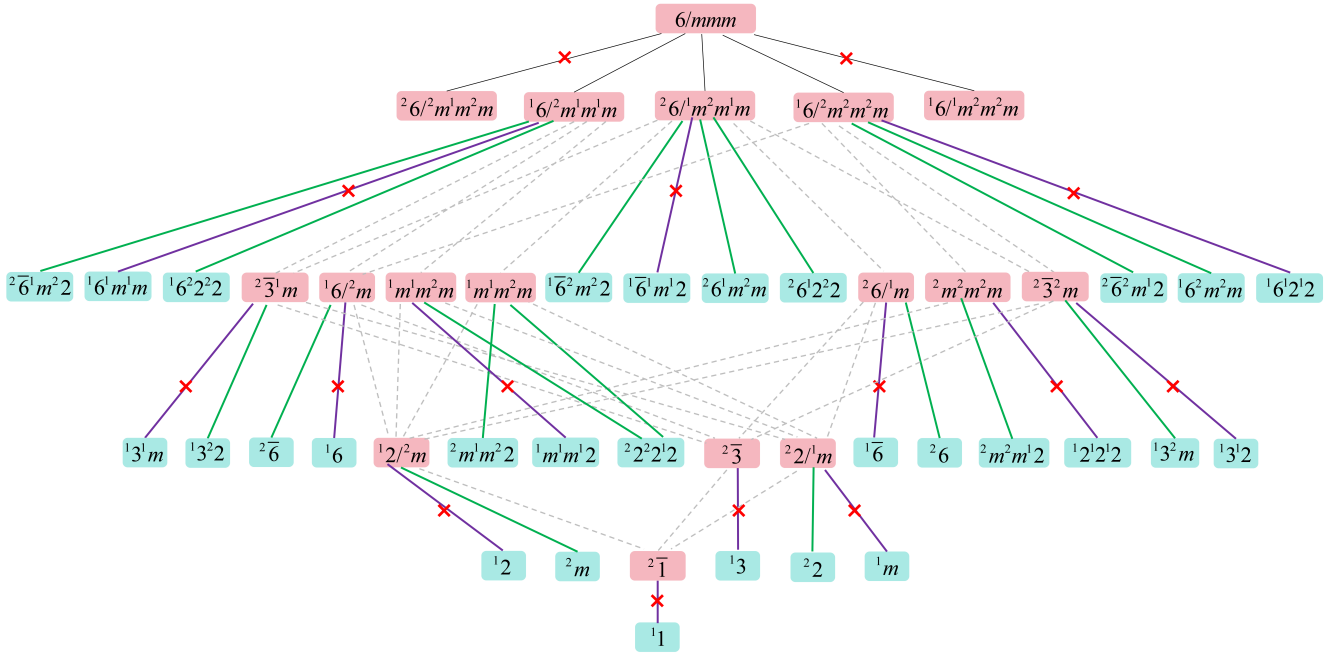


Fig. S8: The spin group evolution of the crystallographic point group $6/mmm$.

9. *The crystallographic point group* $6/mmm$

The parent crystallographic point group $6/mmm$ gives rise to five spin point groups: ${}^{2}6/{}^{2}m^{1}m^{2}m$, ${}^{1}6/{}^{1}m^{2}m^{2}m$, ${}^{2}6/{}^{1}m^{2}m^{1}m$, ${}^{1}6/{}^{2}m^{1}m^{1}m$, and ${}^{1}6/{}^{2}m^{2}m^{2}m$. Among these, ${}^{2}6/{}^{2}m^{1}m^{2}m$ and ${}^{1}6/{}^{1}m^{2}m^{2}m$ describe altermagnetic configurations. The remaining three groups, ${}^{2}6/{}^{1}m^{2}m^{1}m$, ${}^{1}6/{}^{2}m^{1}m^{1}m$, and ${}^{1}6/{}^{2}m^{2}m^{2}m$, contain inversion symmetry P in their cosets and thus characterize conventional antiferromagnetic phases.

These antiferromagnetic spin groups can undergo various symmetry breakings to transition into altermagnetic phases. Starting with ${}^{2}6/{}^{1}m^{2}m^{1}m$, breaking $[C_2\|P]$, $[C_2\|C_6]$, $[C_2\|C_3]$, and the three vertical mirrors leads to ${}^{1}\bar{6}^{2}m^{2}2$. Alternatively, breaking $[C_2\|P]$, $[C_2\|C_3]$, $[C_2\|C_{2x}]$, $[E\|C_6]$, and the three in-plane C_2 rotations results in ${}^{2}6^{1}m^{2}m$, while the combination of $[C_2\|P]$, $[C_2\|C_3]$, three vertical mirrors, and three in-plane C_2 symmetries gives rise to ${}^{2}6^{1}2^{2}2$. This spin group can also transition via intermediate subgroups such as ${}^{2}\bar{3}^{1}m$, ${}^{2}\bar{3}^{2}m$, ${}^{2}6/{}^{1}m$, ${}^{1}m^{1}m^{2}m$, ${}^{1}2/{}^{2}m$, and ${}^{2}2/{}^{1}m$, each retaining inversion symmetry, with subsequent breaking of P enabling access to altermagnetic states. In total, ${}^{2}6/{}^{1}m^{2}m^{1}m$ supports 10 distinct symmetry-breaking pathways.

For ${}^{1}6/{}^{2}m^{1}m^{1}m$, breaking $[C_2\|P]$, $[C_2\|C_3]$, $[C_2\|C_{2\underline{x}}]$, $[E\|C_6]$, and the three vertical mirrors leads to the altermagnetic group ${}^{2}\bar{6}^{1}m^{2}2$, while breaking $[C_2\|P]$, $[C_2\|\bar{C}_3]$, $[C_2\|\bar{C}_6]$, and all vertical mirrors yields ${}^{1}6^{2}2^{2}2$. Indirect transitions through subgroups such as ${}^{2}\bar{3}^{1}m$, ${}^{1}6/{}^{2}m$, ${}^{1}m^{1}m^{2}m$, ${}^{1}2/{}^{2}m$, and ${}^{2}2/{}^{1}m$ are also possible, with inversion symmetry breaking at a later stage. This group thus admits seven distinct symmetry-breaking routes to altermagnetism.

Similarly, for ${}^{1}6/{}^{2}m^{2}m^{2}m$, breaking $[C_2\|P]$, $[C_2\|C_3]$, $[C_2\|M_x]$, $[E\|C_6]$, and three in-plane C_2 symmetries leads to the altermagnetic group ${}^{2}\bar{6}^{1}2^{2}m$, while breaking $[C_2\|P]$, $[C_2\|C_3]$, $[C_2\|C_6]$, and all in-plane C_2 operations yields ${}^{1}6^{2}m^{2}m$. This group can also transition via intermediate subgroups such as ${}^{2}\bar{3}^{2}m$, ${}^{1}6/{}^{2}m$, ${}^{2}m^{2}m^{2}m$, and ${}^{1}2/{}^{2}m$, followed by inversion symmetry breaking. Altogether, ${}^{1}6/{}^{2}m^{2}m^{2}m$ permits six symmetry-breaking paths.

In summary, the three antiferromagnetic spin groups ${}^{2}6/{}^{1}m^{2}m^{1}m$, ${}^{1}6/{}^{2}m^{1}m^{1}m$, and ${}^{1}6/{}^{2}m^{2}m^{2}m$ offer a total of 23 distinct symmetry-breaking transitions connecting antiferromagnetic and altermagnetic phases, as summarized in Fig. S8 and Tab. S2.

10. The crystallographic point group $m\bar{3}$

The parent crystallographic point group $m\bar{3}$ gives rise to the spin point group ${}^{2}m^{2}\bar{3}$, which corresponds to antiferromagnetic phases. The spin group ${}^{2}m^{2}\bar{3}$ cannot directly transition into an altermagnetic phase through inversion symmetry breaking alone, as illustrated in Fig. S9. Instead, two symmetry-breaking pathways are required. One involves breaking the $C_{3_{111}}$ symmetry, reducing the system to the antiferromagnetic spin group ${}^{2}m^{2}m^{2}m$, followed by inversion symmetry breaking to yield the altermagnetic group ${}^{2}m^{2}m^{1}2$. Alternatively, breaking the $C_{3_{111}}$ symmetry along with rotational or mirror symmetries while preserving inversion leads to the intermediate antiferromagnetic group ${}^{1}2/{}^{2}m$, which can subsequently break inversion to transition into the altermagnetic group ${}^{2}m$. In summary, the point group $m\bar{3}$ supports two distinct symmetry-breaking routes to an altermagnetic phase, as detailed in Fig. S9 and Tab. S2.

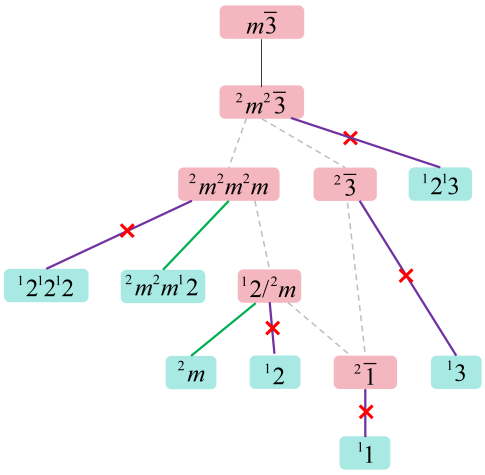


Fig. S9: The spin group evolution of the crystallographic point group $m\bar{3}$.

11. *The crystallographic point group $m\bar{3}m$*

The parent crystallographic point group $m\bar{3}m$ gives rise to three spin point groups: ${}^{1}m{}^{1}\bar{3}{}^{2}m$, ${}^{2}m{}^{2}\bar{3}{}^{1}m$, and ${}^{2}m{}^{2}\bar{3}{}^{2}m$. Among them, ${}^{1}m{}^{1}\bar{3}{}^{2}m$ describe altermagnetic states, while ${}^{2}m{}^{2}\bar{3}{}^{1}m$ and ${}^{2}m{}^{2}\bar{3}{}^{2}m$, both containing inversion symmetry P in their cosets, correspond to conventional antiferromagnetic phases. For ${}^{2}m{}^{2}\bar{3}{}^{1}m$, breaking the symmetries $[C_2\|P]$, $[C_2\|\bar{C}_{3_{111}}]$, $[C_2\|M_{x/y/z}]$, along with all mirror planes and fourfold roto-inversion elements, leads to the altermagnetic spin group ${}^{2}4{}^{1}3{}^{2}2$. Similarly, in the case of ${}^{2}m{}^{2}\bar{3}{}^{2}m$, breaking $[C_2\|P]$, $[C_2\|\bar{C}_{3_{111}}]$, $[C_2\|M_{x/y/z}]$, and all fourfold rotations gives rise to the altermagnetic spin group ${}^{2}\bar{4}{}^{1}3{}^{2}m$. Both antiferromagnetic spin groups can also undergo a series of symmetry reductions to intermediate subgroups, such as ${}^{2}4/{}^{2}m{}^{2}m{}^{1}m$, ${}^{1}4/{}^{2}m{}^{2}m{}^{2}m$, ${}^{2}\bar{3}{}^{1}m$, ${}^{2}\bar{3}{}^{2}m$, ${}^{2}m{}^{2}\bar{3}$, ${}^{2}4/{}^{2}m$, ${}^{1}4/{}^{2}m$, ${}^{1}m{}^{1}m{}^{2}m$, ${}^{2}m{}^{2}m{}^{2}m$, ${}^{2}2/{}^{1}m$, and ${}^{1}2/{}^{2}m$, which retain inversion symmetry. Subsequent breaking of P in these subgroups enables the transition into altermagnetic phases, following symmetry-breaking mechanisms established for point groups such as $4/mmm$, $\bar{3}m$, $m\bar{3}$, $4/m$, mmm, and $2/m$. In total, the $m\bar{3}m$ point group permits 17 distinct symmetry-breaking pathways from antiferromagnetism to altermagnetism, as summarized in Fig. S10 and Tab. S2.

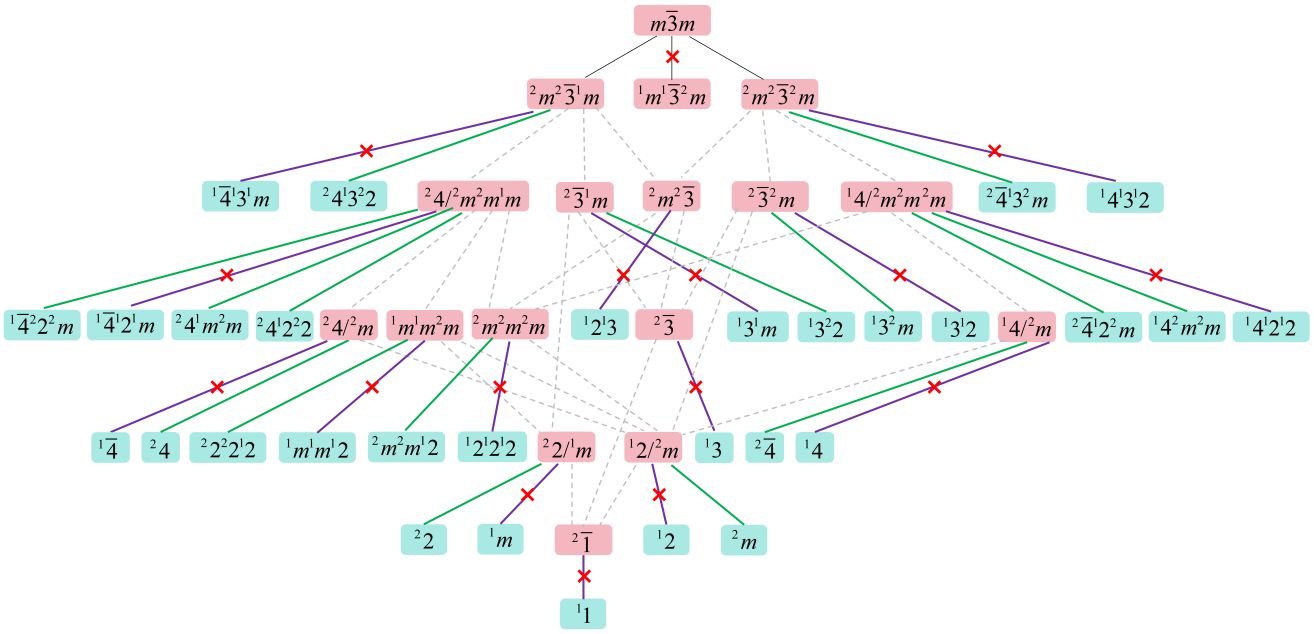


Fig. S10: The spin group evolution of the crystallographic point group $m\bar{3}m$.

Tab. S1: There are 58 spin point groups describing collinear magnets with antiferromagnetic coupling in real space. The symbols $\mathcal{G}$, $\mathcal{N}$, and $\mathcal{R}_s$ denote the crystallographic point group, its halving subgroups, and the spin point group, respectively. The group $\mathcal{B}$ is defined as a group isomorphic to the factor group $\mathcal{G}/\mathcal{N}$. Symmetry operations and magnetic point groups are represented by SO and MPG, respectively. The spin point groups $\mathcal{R}_s$ are highlighted in light red and green, corresponding to antiferromagnetic and altermagnetic phases in collinear magnets, respectively. Additionally, the use of red in symmetry operations indicates that the invariant subgroup corresponds to the symmetry within the coset $\mathcal{A}\mathcal{N}$ that simultaneously acts on both real space and spin space.

$\mathcal{G}$	$\mathcal{N}$	$\mathcal{B}$	$\mathcal{R}_s$	SO	MPG
$\bar{1}$ ($\mathcal{C}_i$)	1	2	${}^{2}\bar{1}$	$[E\|\|E]$,$[C_2\|\|\mathcal{P}]$	$\bar{1}'$
2 ($\mathcal{C}_2$)	1	2	${}^{2}2$	$[E\|\|E]$,$[C_2\|\|C_2]$	$2'$
m ($\mathcal{C}_s$)	1	2	${}^{2}m$	$[E\|\|E]$,$[C_2\|\|M]$	m'
$2/m$ ($\mathcal{C}_{2h}$)	2	2	${}^{1}2/{}^{2}m$	$[E\|\|E]$,$[E\|\|C_2]$,$[C_2\|\|\mathcal{P}]$,$[C_2\|\|M]$	$2/m'$
	m	2	${}^{2}2/{}^{1}m$	$[E\|\|E]$,$[E\|\|M]$,$[C_2\|\|C_2]$,$[C_2\|\|\mathcal{P}]$	$2'/m$
	$\bar{1}$	2	${}^{2}2/{}^{2}m$	$[E\|\|E]$,$[E\|\|\mathcal{P}]$,$[C_2\|\|C_2]$,$[C_2\|\|M]$	$2'/m'$
222 ($\mathcal{D}_2$)	2	2	${}^{2}2{}^{2}2{}^{1}2$	$[E\|\|E]$,$[E\|\|C_{2z}]$,$[C_2\|\|C_{2x}]$,$[C_2\|\|C_{2y}]$	$2'2'2$
$mm2$ ($\mathcal{C}_{2v}$)	2	2	${}^{2}m{}^{2}m{}^{1}2$	$[E\|\|E]$,$[E\|\|C_{2z}]$,$[C_2\|\|M_x]$,$[C_2\|\|M_y]$	$m'm'2$
	m	2	${}^{2}m{}^{1}m{}^{2}2$	$[E\|\|E]$,$[E\|\|M_y]$,$[C_2\|\|M_x]$,$[C_2\|\|C_{2z}]$	$m'm2'$
mmm ($\mathcal{D}_{2h}$)	$2/m$	2	${}^{2}m{}^{2}m{}^{1}m$	$[E\|\|E]$,$[E\|\|C_{2z}]$,$[E\|\|\mathcal{P}]$,$[E\|\|M_z]$, $[C_2\|\|M_x]$,$[C_2\|\|M_y]$,$[C_2\|\|C_{2x}]$,$[C_2\|\|C_{2y}]$	$m'm'm$
	$mm2$	2	${}^{1}m{}^{1}m{}^{2}m$	$[E\|\|E]$,$[E\|\|M_x]$,$[E\|\|M_y]$,$[E\|\|C_{2z}]$, $[C_2\|\|C_{2x}]$,$[C_2\|\|C_{2y}]$,$[C_2\|\|\mathcal{P}]$,$[C_2\|\|M_z]$	mmm'
	222	2	${}^{2}m{}^{2}m{}^{2}m$	$[E\|\|E]$,$[E\|\|C_{2x}]$,$[E\|\|C_{2y}]$,$[E\|\|C_{2z}]$, $[C_2\|\|M_x]$,$[C_2\|\|M_y]$,$[C_2\|\|M_z]$,$[C_2\|\|\mathcal{P}]$	$m'm'm'$
4 ($\mathcal{C}_4$)	2	2	${}^{2}4$	$[E\|\|E]$,$[E\|\|C_{2z}]$,$[C_2\|\|C_{4z}^{+}]$,$[C_2\|\|C_{4z}^{-}]$	$4'$
$\bar{4}$ ($\mathcal{S}_4$)	2	2	${}^{2}\bar{4}$	$[E\|\|E]$,$[E\|\|C_{2z}]$,$[C_2\|\|\bar{C}_{4z}^{+}]$,$[C_2\|\|\bar{C}_{4z}^{-}]$	$\bar{4}'$
$4/m$ ($\mathcal{C}_{4h}$)	$2/m$	2	${}^{2}4/{}^{1}m$	$[E\|\|E]$,$[E\|\|C_{2z}]$,$[E\|\|\mathcal{P}]$,$[E\|\|M_z]$, $[C_2\|\|C_{4z}^{+}]$,$[C_2\|\|C_{4z}^{-}]$,$[C_2\|\|\bar{C}_{4z}^{+}]$,$[C_2\|\|\bar{C}_{4z}^{-}]$	$4'/m$
	$\bar{4}$	2	${}^{2}4/{}^{2}m$	$[E\|\|E]$,$[E\|\|C_{2z}]$,$[E\|\|\bar{C}_{4z}^{+}]$,$[E\|\|\bar{C}_{4z}^{-}]$, $[C_2\|\|C_{4z}^{+}]$,$[C_2\|\|C_{4z}^{-}]$,$[C_2\|\|\mathcal{P}]$,$[C_2\|\|M_z]$	$4'/m'$
	4	2	${}^{1}4/{}^{2}m$	$[E\|\|E]$,$[E\|\|C_{2z}]$,$[E\|\|C_{4z}^{+}]$,$[E\|\|C_{4z}^{-}]$, $[C_2\|\|\bar{C}_{4z}^{+}]$,$[C_2\|\|\bar{C}_{4z}^{-}]$,$[C_2\|\|\mathcal{P}]$,$[C_2\|\|M_z]$	$4/m'$
422 ($\mathcal{D}_4$)	4	2	${}^{1}4{}^{2}2{}^{2}2$	$[E\|\|E]$,$[E\|\|C_{2z}]$,$[E\|\|C_{4z}^{+}]$,$[E\|\|C_{4z}^{-}]$, $[C_2\|\|C_{2x}]$,$[C_2\|\|C_{2y}]$,$[C_2\|\|C_{2xy}]$,$[C_2\|\|C_{2\bar{x}y}]$	$42'2'$
	222	2	${}^{2}4{}^{1}2{}^{2}2$	$[E\|\|E]$,$[E\|\|C_{2z}]$,$[E\|\|C_{2x}]$,$[E\|\|C_{2y}]$, $[C_2\|\|C_{4z}^{+}]$,$[C_2\|\|C_{4z}^{-}]$,$[C_2\|\|C_{2xy}]$,$[C_2\|\|C_{2\bar{x}y}]$	$4'22'$
$4mm$ ($\mathcal{C}_{4v}$)	4	2	${}^{1}4{}^{2}m{}^{2}m$	$[E\|\|E]$,$[E\|\|C_{2z}]$,$[E\|\|C_{4z}^{+}]$,$[E\|\|C_{4z}^{-}]$, $[C_2\|\|M_x]$,$[C_2\|\|M_y]$,$[C_2\|\|M_{xy}]$,$[C_2\|\|M_{\bar{x}y}]$	$4m'm'$
	$mm2$	2	${}^{2}4{}^{1}m{}^{2}m$	$[E\|\|E]$,$[E\|\|C_{2z}]$,$[E\|\|M_x]$,$[E\|\|M_y]$, $[C_2\|\|C_{4z}^{+}]$,$[C_2\|\|C_{4z}^{-}]$,$[C_2\|\|M_{xy}]$,$[C_2\|\|M_{\bar{x}y}]$	$4'mm'$

$\mathcal{G}$	$\mathcal{N}$	$\mathcal{B}$	$\mathcal{R}_s$	SO	MPG
$\bar{4}2m$ ($\mathcal{D}_{2d}$)	$\bar{4}$	2	${}^{1}\bar{4}^{2}2^{2}m$	$[E\|\|E],[E\|\|C_{2z}],[E\|\|\bar{C}^{+}_{4z}],[E\|\|\bar{C}^{-}_{4z}]$, $[C_2\|\|C_{2x}],[C_2\|\|C_{2y}],[C_2\|\|M_{xy}],[C_2\|\|M_{\bar{x}y}]$	$\bar{4}2'm'$
	$mm2$	2	${}^{2}\bar{4}^{2}2^{1}m$	$[E\|\|E],[E\|\|C_{2z}],[E\|\|M_{xy}],[E\|\|M_{\bar{x}y}]$, $[C_2\|\|\bar{C}^{+}_{4z}],[C_2\|\|\bar{C}^{-}_{4z}],[C_2\|\|C_{2x}],[C_2\|\|C_{2y}]$	$\bar{4}'2'm$
	222	2	${}^{2}\bar{4}^{1}2^{2}m$	$[E\|\|E],[E\|\|C_{2z}],[E\|\|C_{2x}],[E\|\|C_{2y}]$, $[C_2\|\|\bar{C}^{+}_{4z}],[C_2\|\|\bar{C}^{-}_{4z}],[C_2\|\|M_{xy}],[C_2\|\|M_{\bar{x}y}]$	$\bar{4}'2m'$
$4/mmm$ ($\mathcal{D}_{4h}$)	$\bar{4}2m$	2	${}^{2}4/{}^{2}m^{2}m^{1}m$	$[E\|\|E],[E\|\|C_{2z}],[E\|\|\bar{C}^{+}_{4z}],[E\|\|\bar{C}^{-}_{4z}],[E\|\|C_{2x}],[E\|\|C_{2y}]$, $[E\|\|M_{xy}],[E\|\|M_{\bar{x}y}],[C_2\|\|\mathcal{P}],[C_2\|\|M_z],[C_2\|\|C^{+}_{4z}],[C_2\|\|C^{-}_{4z}]$, $[C_2\|\|M_x],[C_2\|\|M_y],[C_2\|\|C_{2xy}],[C_2\|\|C_{2\bar{x}y}]$	$4'/m'm'm$
	$4mm$	2	${}^{1}4/{}^{2}m^{1}m^{1}m$	$[E\|\|E],[E\|\|C_{2z}],[E\|\|C^{+}_{4z}],[E\|\|C^{-}_{4z}],[E\|\|M_x],[E\|\|M_y]$, $[E\|\|M_{xy}],[E\|\|M_{\bar{x}y}],[C_2\|\|\mathcal{P}],[C_2\|\|M_z],[C_2\|\|\bar{C}^{+}_{4z}],[C_2\|\|\bar{C}^{-}_{4z}]$, $[C_2\|\|C_{2x}],[C_2\|\|C_{2y}],[C_2\|\|C_{2xy}],[C_2\|\|C_{2\bar{x}y}]$	$4/m'mm$
	mmm	2	${}^{2}4/{}^{1}m^{1}m^{2}m$	$[E\|\|E],[E\|\|C_{2z}],[E\|\|C_{2x}],[E\|\|C_{2y}],[E\|\|\mathcal{P}],[E\|\|M_z]$, $[E\|\|M_x],[E\|\|M_y],[C_2\|\|C^{+}_{4z}],[C_2\|\|C^{-}_{4z}],[C_2\|\|C_{2xy}],[C_2\|\|C_{2\bar{x}y}]$, $[C_2\|\|\bar{C}^{+}_{4z}],[C_2\|\|\bar{C}^{-}_{4z}],[C_2\|\|M_{xy}],[C_2\|\|M_{\bar{x}y}]$	$4'/mmm'$
	$4/m$	2	${}^{1}4/{}^{1}m^{2}m^{2}m$	$[E\|\|E],[E\|\|C_{2z}],[E\|\|C^{+}_{4z}],[E\|\|C^{-}_{4z}],[E\|\|\mathcal{P}],[E\|\|M_z]$, $[E\|\|\bar{C}^{+}_{4z}],[E\|\|\bar{C}^{-}_{4z}],[C_2\|\|C_{2x}],[C_2\|\|C_{2y}],[C_2\|\|C_{2xy}],[C_2\|\|C_{2\bar{x}y}]$, $[C_2\|\|M_x],[C_2\|\|M_y],[C_2\|\|M_{xy}],[C_2\|\|M_{\bar{x}y}]$	$4/mm'm'$
	422	2	${}^{1}4/{}^{2}m^{2}m^{2}m$	$[E\|\|E],[E\|\|C_{2z}],[E\|\|C^{+}_{4z}],[E\|\|C^{-}_{4z}],[E\|\|C_{2x}],[E\|\|C_{2y}]$, $[E\|\|C_{2xy}],[E\|\|C_{2\bar{x}y}],[C_2\|\|\mathcal{P}],[C_2\|\|M_z],[C_2\|\|\bar{C}^{+}_{4z}],[C_2\|\|\bar{C}^{-}_{4z}]$, $[C_2\|\|M_x],[C_2\|\|M_y],[C_2\|\|M_{xy}],[C_2\|\|M_{\bar{x}y}]$	$4/m'm'm'$
$\bar{3}$ ($\mathcal{C}_{3i}$)	3	2	${}^{2}\bar{3}$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[C_2\|\|\mathcal{P}],[C_2\|\|\bar{C}^{+}_{3z}],[C_2\|\|\bar{C}^{-}_{3z}]$	$\bar{3}'$
32 ($\mathcal{D}_3$)	3	2	${}^{1}3^{2}2$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[C_2\|\|C_{2x}],[C_2\|\|C_{2y}],[C_2\|\|C_{2xy}]$	$32'$
$3m$ ($\mathcal{C}_{3v}$)	3	2	${}^{1}3^{2}m$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[C_2\|\|M_x],[C_2\|\|M_y],[C_2\|\|M_{xy}]$	$3m'$
$\bar{3}m$ ($\mathcal{D}_{3d}$)	$\bar{3}$	2	${}^{1}\bar{3}^{2}m$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[E\|\|\mathcal{P}],[E\|\|\bar{C}^{+}_{3z}],[E\|\|\bar{C}^{-}_{3z}]$, $[C_2\|\|C_{2x}],[C_2\|\|C_{2y}],[C_2\|\|C_{2xy}],[C_2\|\|M_x],[C_2\|\|M_y],[C_2\|\|M_{xy}]$	$\bar{3}m'$
	$3m$	2	${}^{2}\bar{3}^{1}m$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[E\|\|M_x],[E\|\|M_y],[E\|\|M_{xy}]$, $[C_2\|\|\mathcal{P}],[C_2\|\|\bar{C}^{+}_{3z}],[C_2\|\|\bar{C}^{-}_{3z}],[C_2\|\|C_{2x}],[C_2\|\|C_{2y}],[C_2\|\|C_{2xy}]$	$\bar{3}'m$
	32	2	${}^{2}\bar{3}^{2}m$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[E\|\|C_{2x}],[E\|\|C_{2y}],[E\|\|C_{2xy}]$, $[C_2\|\|\mathcal{P}],[C_2\|\|\bar{C}^{+}_{3z}],[C_2\|\|\bar{C}^{-}_{3z}],[C_2\|\|M_x],[C_2\|\|M_y],[C_2\|\|M_{xy}]$	$\bar{3}'m'$
6 ($\mathcal{C}_6$)	3	2	${}^{2}6$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[C_2\|\|C_{2z}],[C_2\|\|C^{+}_{6z}],[C_2\|\|C^{-}_{6z}]$	$6'$
$\bar{6}$ ($\mathcal{C}_{3h}$)	3	2	${}^{2}\bar{6}$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[C_2\|\|M_z],[C_2\|\|\bar{C}^{+}_{6z}],[C_2\|\|\bar{C}^{-}_{6z}]$	$\bar{6}'$
$6/m$ ($\mathcal{C}_{6h}$)	$\bar{3}$	2	${}^{2}6/{}^{2}m$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[E\|\|\bar{C}^{+}_{3z}],[E\|\|\bar{C}^{-}_{3z}],[E\|\|\mathcal{P}]$, $[C_2\|\|C_{2z}],[C_2\|\|C^{+}_{6z}],[C_2\|\|C^{-}_{6z}],[C_2\|\|M_z],[C_2\|\|\bar{C}^{+}_{6z}],[C_2\|\|\bar{C}^{-}_{6z}]$	$6'/m'$
	$\bar{6}$	2	${}^{2}6/{}^{1}m$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[E\|\|M_z],[E\|\|\bar{C}^{+}_{6z}],[E\|\|\bar{C}^{-}_{6z}]$, $[C_2\|\|C_{2z}],[C_2\|\|C^{+}_{6z}],[C_2\|\|C^{-}_{6z}],[C_2\|\|\mathcal{P}],[C_2\|\|\bar{C}^{+}_{3z}],[C_2\|\|\bar{C}^{-}_{3z}]$	$6'/m$
	6	2	${}^{1}6/{}^{2}m$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[E\|\|C_{2z}],[E\|\|C^{+}_{6z}],[E\|\|C^{-}_{6z}]$, $[C_2\|\|\mathcal{P}],[C_2\|\|\bar{C}^{+}_{3z}],[C_2\|\|\bar{C}^{-}_{3z}],[C_2\|\|M_z],[C_2\|\|\bar{C}^{+}_{6z}],[C_2\|\|\bar{C}^{-}_{6z}]$	$6/m'$
622 ($\mathcal{D}_6$)	6	2	${}^{1}6^{2}2^{2}2$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[E\|\|C_{2z}],[E\|\|C^{+}_{6z}],[E\|\|C^{-}_{6z}]$, $[C_2\|\|C_{2x}],[C_2\|\|C_{2y}],[C_2\|\|C_{2xy}],[C_2\|\|C_{21}],[C_2\|\|C_{22}],[C_2\|\|C_{23}]$	$62'2'$
	32	2	${}^{2}6^{1}2^{2}2$	$[E\|\|E],[E\|\|C^{+}_{3z}],[E\|\|C^{-}_{3z}],[E\|\|C_{2x}],[E\|\|C_{2y}],[E\|\|\|C_{2xy}]$, $[C_2\|\|C_{2z}],[C_2\|\|C^{+}_{6z}],[C_2\|\|C^{-}_{6z}],[C_2\|\|C_{21}],[C_2\|\|C_{22}],[C_2\|\|C_{23}]$	$6'22'$

$\mathcal{G}$	$\mathcal{N}$	$\mathcal{B}$	$\mathcal{R}_s$	SO	MPG
$6mm$ ($\mathcal{C}_{6v}$)	6	2	${}^{1}6^{2}m^{2}m$	$[E\|\|E],[E\|\|C_{3z}^{+}],[E\|\|C_{3z}^{-}],[E\|\|C_{2z}],[E\|\|C_{6z}^{+}],[E\|\|C_{6z}^{-}]$, $[C_2\|\|M_x],[C_2\|\|M_y],[C_2\|\|M_{xy}],[C_2\|\|M_1],[C_2\|\|M_2],[C_2\|\|M_3]$	$6m'm'$
	$3m$	2	${}^{2}6^{1}m^{2}m$	$[E\|\|E],[E\|\|C_{3z}^{+}],[E\|\|C_{3z}^{-}],[[E\|\|M_x],[E\|\|M_y],[E\|\|M_{xy}]$, $[C_2\|\|C_{2z}],[C_2\|\|C_{6z}^{+}],[C_2\|\|C_{6z}^{-}],[C_2\|\|M_1],[C_2\|\|M_2],[C_2\|\|M_3]$	$6'mm'$
$\bar{6}m2$ ($\mathcal{D}_{3h}$)	$\bar{6}$	2	${}^{1}\bar{6}^{2}m^{2}2$	$[E\|\|E],[E\|\|C_{3z}^{+}],[E\|\|C_{3z}^{-}],[E\|\|M_z],[E\|\|\bar{C}_{6z}^{+}],[E\|\|\bar{C}_{6z}^{-}]$, $[C_2\|\|M_x],[C_2\|\|M_y],[C_2\|\|M_{xy}],[C_2\|\|C_{2_1}],[C_2\|\|C_{2_2}],[C_2\|\|C_{2_3}]$	$\bar{6}m'2'$
	$3m$	2	${}^{2}\bar{6}^{1}m^{2}2$	$[E\|\|E],[E\|\|C_{3z}^{+}],[E\|\|C_{3z}^{-}],[[E\|\|M_x],[E\|\|M_y],[E\|\|M_{xy}]$, $[C_2\|\|M_z],[C_2\|\|\bar{C}_{6z}^{+}],[C_2\|\|\bar{C}_{6z}^{-}],[C_2\|\|C_{2_1}],[C_2\|\|C_{2_2}],[C_2\|\|C_{2_3}]$	$\bar{6}'m2'$
	32	2	${}^{2}\bar{6}^{2}m^{1}2$	$[E\|\|E],[E\|\|C_{3z}^{+}],[E\|\|C_{3z}^{-}],[[E\|\|C_{2_1}],[E\|\|C_{2_2}],[E\|\|C_{2_3}]$, $[C_2\|\|M_z],[C_2\|\|\bar{C}_{6z}^{+}],[C_2\|\|\bar{C}_{6z}^{-}],[C_2\|\|M_x],[C_2\|\|M_y],[C_2\|\|M_{xy}]$	$\bar{6}'m'2$
$6/mmm$ ($\mathcal{D}_{6h}$)	$\bar{3}m$	2	${}^{2}6/{}^{2}m^{1}m^{2}m$	$[E\|\|E],[E\|\|C_{3z}^{+}],[E\|\|C_{3z}^{-}],[E\|\|\mathcal{P}],[E\|\|\bar{C}_{3z}^{+}],[E\|\|\bar{C}_{3z}^{-}]$, $[E\|\|M_x],[E\|\|M_y],[E\|\|M_{xy}],[E\|\|C_{2_1}],[E\|\|C_{2_2}],[E\|\|C_{2_3}]$, $[C_2\|\|C_{2z}],[C_2\|\|C_{6z}^{+}],[C_2\|\|C_{6z}^{-}],[C_2\|\|M_z],[C_2\|\|\bar{C}_{6z}^{+}],[C_2\|\|\bar{C}_{6z}^{-}]$, $[C_2\|\|C_{2x}],[C_2\|\|C_{2y}],[C_2\|\|C_{2xy}],[C_2\|\|M_1],[C_2\|\|M_2],[C_2\|\|M_3]$	$6'/m'mm'$
	$\bar{6}m2$	2	${}^{2}6/{}^{1}m^{2}m^{1}m$	$[E\|\|E],[E\|\|C_{3z}^{+}],[E\|\|C_{3z}^{-}],[E\|\|M_z],[E\|\|\bar{C}_{6z}^{+}],[E\|\|\bar{C}_{6z}^{-}]$, $[E\|\|M_x],[E\|\|M_y],[E\|\|M_{xy}],[E\|\|C_{2_1}],[E\|\|C_{2_2}],[E\|\|C_{2_3}]$, $[C_2\|\|\mathcal{P}],[C_2\|\|\bar{C}_{3z}^{+}],[C_2\|\|\bar{C}_{3z}^{-}],[C_2\|\|C_{2z}],[C_2\|\|C_{6z}^{+}],[C_2\|\|C_{6z}^{-}]$, $[C_2\|\|C_{2x}],[C_2\|\|C_{2y}],[C_2\|\|C_{2xy}],[C_2\|\|M_1],[C_2\|\|M_2],[C_2\|\|M_3]$	$6'/mm'm$
	$6/m$	2	${}^{1}6/{}^{1}m^{2}m^{2}m$	$[E\|\|E],[E\|\|C_{3z}^{+}],[E\|\|C_{3z}^{-}],[E\|\|C_{2z}],[E\|\|C_{6z}^{+}],[E\|\|C_{6z}^{-}]$, $[E\|\|\mathcal{P}],[E\|\|\bar{C}_{3z}^{+}],[E\|\|\bar{C}_{3z}^{-}],[E\|\|M_z],[E\|\|\bar{C}_{6z}^{+}],[E\|\|\bar{C}_{6z}^{-}]$, $[C_2\|\|M_x],[C_2\|\|M_y],[C_2\|\|M_{xy}],[C_2\|\|M_1],[C_2\|\|M_2],[C_2\|\|M_3]$, $[C_2\|\|C_{2x}],[C_2\|\|C_{2y}],[C_2\|\|C_{2xy}],[C_2\|\|C_{2_1}],[C_2\|\|C_{2_2}],[C_2\|\|C_{2_3}]$	$6/mm'm'$
	$6mm$	2	${}^{1}6/{}^{2}m^{1}m^{1}m$	$[E\|\|E],[E\|\|C_{3z}^{+}],[E\|\|C_{3z}^{-}],[E\|\|C_{2z}],[E\|\|C_{6z}^{+}],[E\|\|C_{6z}^{-}]$, $[E\|\|M_x],[E\|\|M_y],[E\|\|M_{xy}],[E\|\|M_1],[E\|\|M_2],[E\|\|M_3]$, $[C_2\|\|\mathcal{P}],[C_2\|\|\bar{C}_{3z}^{+}],[C_2\|\|\bar{C}_{3z}^{-}],[C_2\|\|M_z],[C_2\|\|\bar{C}_{6z}^{+}],[C_2\|\|\bar{C}_{6z}^{-}]$, $[C_2\|\|C_{2x}],[C_2\|\|C_{2y}],[C_2\|\|C_{2xy}],[C_2\|\|C_{2_1}],[C_2\|\|C_{2_2}],[C_2\|\|C_{2_3}]$	$6/m'mm$
	622	2	${}^{1}6/{}^{2}m^{2}m^{2}m$	$[E\|\|E],[E\|\|C_{3z}^{+}],[E\|\|C_{3z}^{-}],[E\|\|C_{2z}],[E\|\|C_{6z}^{+}],[E\|\|C_{6z}^{-}]$, $[E\|\|C_{2x}],[E\|\|C_{2y}],[E\|\|C_{2xy}],[E\|\|C_{2_1}],[E\|\|C_{2_2}],[E\|\|C_{2_3}]$, $[C_2\|\|\mathcal{P}],[C_2\|\|\bar{C}_{3z}^{+}],[C_2\|\|\bar{C}_{3z}^{-}],[C_2\|\|M_z],[C_2\|\|\bar{C}_{6z}^{+}],[C_2\|\|\bar{C}_{6z}^{-}]$, $[C_2\|\|M_x],[C_2\|\|M_y],[C_2\|\|M_{xy}],[C_2\|\|M_1],[C_2\|\|M_2],[C_2\|\|M_3]$	$6/m'm'm'$
$m\bar{3}$ ($\mathcal{T}_h$)	23	2	${}^{2}m^{2}\bar{3}$	$[E\|\|E],[E\|\|C_{2x}],[E\|\|C_{2y}],[E\|\|C_{2z}],[E\|\|C_{3_{111}}^{+}],[E\|\|C_{3_{\bar{1}\bar{1}\bar{1}}}^{+}]$, $[E\|\|C_{3_{1\bar{1}\bar{1}}}^{+}],[E\|\|C_{3_{\bar{1}1\bar{1}}}^{+}],[E\|\|C_{3_{11\bar{1}}}^{-}],[E\|\|C_{3_{1\bar{1}\bar{1}}}^{-}],[E\|\|C_{3_{\bar{1}1\bar{1}}}^{-}],[E\|\|C_{3_{\bar{1}\bar{1}1}}^{-}]$, $[C_2\|\|\mathcal{P}],[C_2\|\|M_x],[C_2\|\|M_y],[C_2\|\|M_z],[C_2\|\|\bar{C}_{3_{111}}^{+}],[C_2\|\|\bar{C}_{3_{\bar{1}\bar{1}\bar{1}}}^{+}]$, $[C_2\|\|\bar{C}_{3_{1\bar{1}\bar{1}}}^{+}],[C_2\|\|\bar{C}_{3_{\bar{1}\bar{1}1}}^{+}],[C_2\|\|\bar{C}_{3_{11\bar{1}}}^{-}],[C_2\|\|\bar{C}_{3_{1\bar{1}\bar{1}}}^{-}],[C_2\|\|\bar{C}_{3_{\bar{1}1\bar{1}}}^{-}],[C_2\|\|\bar{C}_{3_{\bar{1}\bar{1}1}}^{-}]$	$m'\bar{3}'$
$\bar{4}3m$ ($\mathcal{T}_d$)	23	2	${}^{2}\bar{4}^{1}3^{2}m$	$[E\|\|E],[E\|\|C_{2x}],[E\|\|C_{2y}],[E\|\|C_{2z}],[E\|\|C_{3_{111}}^{+}],[E\|\|C_{3_{\bar{1}\bar{1}\bar{1}}}^{+}]$, $[E\|\|C_{3_{1\bar{1}\bar{1}}}^{+}],[E\|\|C_{3_{\bar{1}1\bar{1}}}^{+}],[E\|\|C_{3_{11\bar{1}}}^{-}],[E\|\|C_{3_{\bar{1}\bar{1}1}}^{-}],[E\|\|C_{3_{1\bar{1}\bar{1}}}^{-}],[E\|\|C_{3_{\bar{1}1\bar{1}}}^{-}]$, $[C_2\|\|M_{xy}],[C_2\|\|M_{\bar{x}y}],[C_2\|\|M_{yz}],[C_2\|\|M_{\bar{y}z}],[C_2\|\|M_{xz}],[C_2\|\|M_{\bar{x}z}]$, $[C_2\|\|\bar{C}_{4x}^{+}],[C_2\|\|\bar{C}_{4x}^{-}],[C_2\|\|\bar{C}_{4y}^{+}],[C_2\|\|\bar{C}_{4y}^{-}],[C_2\|\|\bar{C}_{4z}^{+}],[C_2\|\|\bar{C}_{4z}^{-}]$	$\bar{4}'3m'$
432 ($\mathcal{O}$)	23	2	${}^{2}4^{1}3^{2}2$	$[E\|\|E],[E\|\|C_{2x}],[E\|\|C_{2y}],[E\|\|C_{2z}],[E\|\|C_{3_{111}}^{+}],[E\|\|C_{3_{\bar{1}\bar{1}\bar{1}}}^{+}]$, $[E\|\|C_{3_{1\bar{1}\bar{1}}}^{+}],[E\|\|C_{3_{\bar{1}1\bar{1}}}^{+}],[E\|\|C_{3_{11\bar{1}}}^{-}],[E\|\|C_{3_{\bar{1}\bar{1}1}}^{-}],[E\|\|C_{3_{1\bar{1}\bar{1}}}^{-}],[E\|\|C_{3_{\bar{1}1\bar{1}}}^{-}]$, $[C_2\|\|C_{2xy}],[C_2\|\|C_{2\bar{x}y}],[C_2\|\|C_{2yz}],[C_2\|\|C_{2\bar{y}z}],[C_2\|\|C_{2xz}],[C_2\|\|C_{2\bar{x}z}]$, $[C_2\|\|C_{4x}^{+}],[C_2\|\|C_{4x}^{-}],[C_2\|\|C_{4y}^{+}],[C_2\|\|C_{4y}^{-}],[C_2\|\|C_{4z}^{+}],[C_2\|\|C_{4z}^{-}]$	$4'32'$

$\mathcal{G}$	$\mathcal{N}$	$\mathcal{B}$	$\mathcal{R}_s$	SO	MPG
$m\bar{3}m$ $(\mathcal{O}_h)$	$m\bar{3}$	2	${}^1m^1\bar{3}^2m$	$[E\|\|E]$,$[E\|\|C_{2x}]$,$[E\|\|C_{2y}]$,$[E\|\|C_{2z}]$,$[E\|\|C^+_{3_{111}}]$,$[E\|\|C^+_{3_{\bar{1}1\bar{1}}}]$, $[E\|\|C^+_{3_{1\bar{1}\bar{1}}}]$,$[E\|\|C^+_{3_{\bar{1}\bar{1}1}}]$,$[E\|\|C^-_{3_{111}}]$,$[E\|\|C^-_{3_{\bar{1}1\bar{1}}}]$,$[E\|\|C^-_{3_{1\bar{1}\bar{1}}}]$,$[E\|\|C^-_{3_{\bar{1}\bar{1}1}}]$, $[E\|\|\mathcal{P}]$,$[E\|\|M_x]$,$[E\|\|M_y]$,$[E\|\|M_z]$,$[E\|\|\bar{C}^+_{3_{111}}]$,$[E\|\|\bar{C}^+_{3_{\bar{1}1\bar{1}}}]$, $[E\|\|\bar{C}^+_{3_{1\bar{1}\bar{1}}}]$,$[E\|\|\bar{C}^+_{3_{\bar{1}\bar{1}1}}]$,$[E\|\|\bar{C}^-_{3_{111}}]$,$[E\|\|\bar{C}^-_{3_{\bar{1}1\bar{1}}}]$,$[E\|\|\bar{C}^-_{3_{1\bar{1}\bar{1}}}]$,$[E\|\|\bar{C}^-_{3_{\bar{1}\bar{1}1}}]$, $[C_2\|\|C_{2xy}]$,$[C_2\|\|C_{2\bar{x}y}]$,$[C_2\|\|C_{2yz}]$,$[C_2\|\|C_{2\bar{y}z}]$,$[C_2\|\|C_{2xz}]$,$[C_2\|\|C_{2\bar{x}z}]$, $[C_2\|\|M_{xy}]$,$[C_2\|\|M_{\bar{x}y}]$,$[C_2\|\|M_{yz}]$,$[C_2\|\|M_{\bar{y}z}]$,$[C_2\|\|M_{xz}]$,$[C_2\|\|M_{\bar{x}z}]$, $[C_2\|\|C^+_{4x}]$,$[C_2\|\|C^-_{4x}]$,$[C_2\|\|C^+_{4y}]$,$[C_2\|\|C^-_{4y}]$,$[C_2\|\|C^+_{4z}]$,$[C_2\|\|C^-_{4z}]$, $[C_2\|\|\bar{C}^+_{4x}]$,$[C_2\|\|\bar{C}^-_{4x}]$,$[C_2\|\|\bar{C}^+_{4y}]$,$[C_2\|\|\bar{C}^-_{4y}]$,$[C_2\|\|\bar{C}^+_{4z}]$,$[C_2\|\|\bar{C}^-_{4z}]$	$m\bar{3}m'$
	$\bar{4}3m$	2	${}^2m^2\bar{3}^1m$	$[E\|\|E]$,$[E\|\|C_{2x}]$,$[E\|\|C_{2y}]$,$[E\|\|C_{2z}]$,$[E\|\|C^+_{3_{111}}]$,$[E\|\|C^+_{3_{\bar{1}1\bar{1}}}]$, $[E\|\|C^+_{3_{1\bar{1}\bar{1}}}]$,$[E\|\|C^+_{3_{\bar{1}\bar{1}1}}]$,$[E\|\|C^-_{3_{111}}]$,$[E\|\|C^-_{3_{\bar{1}1\bar{1}}}]$,$[E\|\|C^-_{3_{1\bar{1}\bar{1}}}]$,$[E\|\|C^-_{3_{\bar{1}\bar{1}1}}]$, $[E\|\|M_{xy}]$,$[E\|\|M_{\bar{x}y}]$,$[E\|\|M_{yz}]$,$[E\|\|M_{\bar{y}z}]$,$[E\|\|M_{xz}]$,$[E\|\|M_{\bar{x}z}]$, $[E\|\|\bar{C}^+_{4x}]$,$[E\|\|\bar{C}^-_{4x}]$,$[E\|\|\bar{C}^+_{4y}]$,$[E\|\|\bar{C}^-_{4y}]$,$[E\|\|\bar{C}^+_{4z}]$,$[E\|\|\bar{C}^-_{4z}]$, $[C_2\|\|\mathcal{P}]$,$[C_2\|\|M_x]$,$[C_2\|\|M_y]$,$[C_2\|\|M_z]$,$[C_2\|\|\bar{C}^+_{3_{111}}]$,$[C_2\|\|\bar{C}^+_{3_{\bar{1}1\bar{1}}}]$, $[C_2\|\|\bar{C}^+_{3_{1\bar{1}\bar{1}}}]$,$[C_2\|\|\bar{C}^+_{3_{\bar{1}\bar{1}1}}]$,$[C_2\|\|\bar{C}^-_{3_{111}}]$,$[C_2\|\|\bar{C}^-_{3_{\bar{1}1\bar{1}}}]$,$[C_2\|\|\bar{C}^-_{3_{1\bar{1}\bar{1}}}]$,$[C_2\|\|\bar{C}^-_{3_{\bar{1}\bar{1}1}}]$, $[C_2\|\|C_{2xy}]$,$[C_2\|\|C_{2\bar{x}y}]$,$[C_2\|\|C_{2yz}]$,$[C_2\|\|C_{2\bar{y}z}]$,$[C_2\|\|C_{2xz}]$,$[C_2\|\|C_{2\bar{x}z}]$, $[C_2\|\|C^+_{4x}]$,$[C_2\|\|C^-_{4x}]$,$[C_2\|\|C^+_{4y}]$,$[C_2\|\|C^-_{4y}]$,$[C_2\|\|C^+_{4z}]$,$[C_2\|\|C^-_{4z}]$	$m'\bar{3}'m$
	432	2	${}^2m^2\bar{3}^2m$	$[E\|\|E]$,$[E\|\|C_{2x}]$,$[E\|\|C_{2y}]$,$[E\|\|C_{2z}]$,$[E\|\|C^+_{3_{111}}]$,$[E\|\|C^+_{3_{\bar{1}1\bar{1}}}]$, $[E\|\|C^+_{3_{1\bar{1}\bar{1}}}]$,$[E\|\|C^+_{3_{\bar{1}\bar{1}1}}]$,$[E\|\|C^-_{3_{111}}]$,$[E\|\|C^-_{3_{\bar{1}1\bar{1}}}]$,$[E\|\|C^-_{3_{1\bar{1}\bar{1}}}]$,$[E\|\|C^-_{3_{\bar{1}\bar{1}1}}]$, $[E\|\|C_{2xy}]$,$[E\|\|C_{2\bar{x}y}]$,$[E\|\|C_{2yz}]$,$[E\|\|C_{2\bar{y}z}]$,$[E\|\|C_{2xz}]$,$[E\|\|C_{2\bar{x}z}]$, $[E\|\|C^+_{4x}]$,$[E\|\|C^-_{4x}]$,$[E\|\|C^+_{4y}]$,$[E\|\|C^-_{4y}]$,$[E\|\|C^+_{4z}]$,$[E\|\|C^-_{4z}]$, $[C_2\|\|M_{xy}]$,$[C_2\|\|M_{\bar{x}y}]$,$[C_2\|\|M_{yz}]$,$[C_2\|\|M_{\bar{y}z}]$,$[C_2\|\|M_{xz}]$,$[C_2\|\|M_{\bar{x}z}]$, $[C_2\|\|\bar{C}^+_{4x}]$,$[C_2\|\|\bar{C}^-_{4x}]$,$[C_2\|\|\bar{C}^+_{4y}]$,$[C_2\|\|\bar{C}^-_{4y}]$,$[C_2\|\|\bar{C}^+_{4z}]$,$[C_2\|\|\bar{C}^-_{4z}]$, $[C_2\|\|\mathcal{P}]$,$[C_2\|\|M_x]$,$[C_2\|\|M_y]$,$[C_2\|\|M_z]$,$[C_2\|\|\bar{C}^+_{3_{111}}]$,$[C_2\|\|\bar{C}^+_{3_{\bar{1}1\bar{1}}}]$, $[C_2\|\|\bar{C}^+_{3_{1\bar{1}\bar{1}}}]$,$[C_2\|\|\bar{C}^+_{3_{\bar{1}\bar{1}1}}]$,$[C_2\|\|\bar{C}^-_{3_{111}}]$,$[C_2\|\|\bar{C}^-_{3_{\bar{1}1\bar{1}}}]$,$[C_2\|\|\bar{C}^-_{3_{1\bar{1}\bar{1}}}]$,$[C_2\|\|\bar{C}^-_{3_{\bar{1}\bar{1}1}}]$	$m'\bar{3}'m'$

Tab. S2: Among the 11 crystallographic point groups with inversion symmetry P in the coset $\mathcal{AN}$, there are 21 distinct spin point groups. A total of 82 pathways exist for magnetic transitions from an antiferromagnetic phase to an altermagnetic phase. Here, $\mathcal{G}$ represents the crystallographic point group, while $\mathcal{R}_s$ denotes the spin point groups. When inversion symmetry is broken, the resulting spin point groups are labeled as $\mathcal{R}_s^{'}$. The check marks highlight the specific pathways through which these magnetic transitions can occur.

$\mathcal{G}$	$\mathcal{R}_s$ (MPG)	$\mathcal{R}_s^{'}$	AM	$\mathcal{G}$	$\mathcal{R}_s$ (MPG)	$\mathcal{R}_s^{'}$	AM
$\bar{1}$ ($\mathcal{C}_i$)	$^2\bar{1}$ ($\bar{1}'$)	$^2\bar{1}$		$4/m$ ($\mathcal{C}_{4h}$)	$^2{4}/^2{m}$ ($4'/m'$)	$^2{4}/^2{m}$	
		$^1{1}$				$^1\bar{4}$	
$2/m$ ($\mathcal{C}_{2h}$)	$^1{2}/^2{m}$ ($2/m'$)	$^1{2}/^2{m}$				$^2{4}$	√
		$^2{m}$	√			$^1{2}/^2{m}$	
		$^1{2}$				$^2{m}$	√
		$^2\bar{1}$				$^1{2}$	
		$^1{1}$				$^2\bar{1}$	
	$^2{2}/^1{m}$ ($2'/m$)	$^2{2}/^1{m}$				$^1{1}$	
		$^1{m}$			$^1{4}/^2{m}$ ($4/m'$)	$^1{4}/^2{m}$	
		$^2{2}$	√			$^2\bar{4}$	√
		$^2\bar{1}$				$^1{4}$	
		$^1{1}$				$^1{2}/^2{m}$	
mmm ($\mathcal{D}_{2h}$)	$^1{m}^1{m}^2{m}$ (mmm')	$^1{m}^1{m}^2{m}$				$^2{m}$	√
		$^2{m}^1{m}^2{2}$	√			$^1{2}$	
		$^1{m}^1{m}^1{2}$				$^2\bar{1}$	
		$^2{2}^2{2}^1{2}$	√			$^1{1}$	
		$^1{2}/^2{m}$		$4/mmm$ ($\mathcal{D}_{4h}$)	$^2{4}/^2{m}^2{m}^1{m}$ ($4'/m'm'm$)	$^2{4}/^2{m}^2{m}^1{m}$	
		$^2{2}/^1{m}$				$^1\bar{4}^2{2}^2{m}$	√
		$^2{m}$	√			$^1\bar{4}^1{2}^1{m}$	
		$^1{m}$				$^2{4}^2{m}^1{m}$	√
		$^2{2}$	√			$^2{4}^1{2}^2{2}$	√
		$^1{2}$				$^2{4}/^2{m}$	
		$^2\bar{1}$				$^1\bar{4}$	
		$^1{1}$				$^2{4}$	√
	$^2{m}^2{m}^2{m}$ ($m'm'm'$)	$^2{m}^2{m}^2{m}$				$^2{m}^2{m}^2{m}$	
		$^2{m}^2{m}^1{2}$	√			$^2{m}^1{m}^1{m}$	
		$^1{2}^1{2}^1{2}$				$^2{m}^2{m}^1{2}$	√
		$^1{2}/^2{m}$				$^1{m}^1{m}^1{2}$	
		$^2{m}$	√			$^2{2}^2{2}^1{2}$	√
		$^1{2}$				$^1{2}^1{2}^1{2}$	
		$^2\bar{1}$				$^1{2}/^2{m}$	
		$^1{1}$				$^2{2}/^1{m}$	

$\mathcal{G}$	$\mathcal{R}_s$ (MPG)	$\mathcal{R}_s^{'}$	AM	$\mathcal{G}$	$\mathcal{R}_s$ (MPG)	$\mathcal{R}_s^{'}$	AM
$4/mmm$ ($\mathcal{D}_{4h}$)	$^{2}4/^{2}m^{2}m^{1}m$ ($4'/m'm'm$)	^{2}m	√	$\bar{3}$ ($\mathcal{C}_{3i}$)	$^{2}\bar{3}$ ($\bar{3}'$)	$^{2}\bar{3}$	
		^{1}m				$^{1}3$	
		$^{2}2$	√			$^{2}\bar{1}$	
		$^{2}\bar{1}$				$^{1}1$	
		$^{1}1$		$\bar{3}m$ ($\mathcal{D}_{3d}$)	$^{2}\bar{3}^{1}m$ ($\bar{3}'m$)	$^{2}\bar{3}^{1}m$	
	$^{1}4/^{2}m^{1}m^{1}m$ ($4/m'mm$)	$^{1}4/^{2}m^{1}m^{1}m$				$^{1}3^{1}m$	
		$^{2}\bar{4}^{2}2^{1}m$	√			$^{1}3^{2}2$	√
		$^{1}4^{1}m^{1}m$				$^{2}\bar{3}$	
		$^{1}4^{2}2^{2}2$	√			$^{1}3$	
		$^{1}4/^{2}m$				$^{2}2/^{1}m$	
		$^{2}\bar{4}$	√			^{1}m	
		$^{1}4$				$^{2}2$	√
		$^{2}m^{1}m^{1}m$				$^{2}\bar{1}$	
		$^{2}m^{1}m^{2}2$	√			$^{1}1$	
		$^{1}m^{1}m^{1}2$			$^{2}\bar{3}^{2}m$ ($\bar{3}'m'$)	$^{2}\bar{3}^{2}m$	
		$^{2}2^{2}2^{1}2$	√			$^{1}3^{2}m$	√
		$^{1}2/^{2}m$				$^{1}3^{1}2$	
		$^{2}2/^{1}m$				$^{2}\bar{3}$	
		^{2}m	√			$^{1}3$	
		^{1}m				$^{1}2/^{2}m$	
		$^{2}2$	√			^{2}m	√
		$^{1}2$				$^{1}2$	
		$^{2}\bar{1}$				$^{2}\bar{1}$	
		$^{1}1$				$^{1}1$	
	$^{1}4/^{2}m^{2}m^{2}m$ ($4/m'm'm'$)	$^{1}4/^{2}m^{2}m^{2}m$		$6/m$ ($\mathcal{C}_{6h}$)	$^{2}6/^{1}m$ ($6'/m$)	$^{2}6/^{1}m$	
		$^{2}\bar{4}^{1}2^{2}m$	√			$^{1}\bar{6}$	
		$^{1}4^{2}m^{2}m$	√			$^{2}6$	√
		$^{1}4^{1}2^{1}2$				$^{2}\bar{3}$	
		$^{1}4/^{2}m$				$^{1}3$	
		$^{2}\bar{4}$	√			$^{2}2/^{1}m$	
		$^{1}4$				^{1}m	
		$^{2}m^{2}m^{2}m$				$^{2}2$	√
		$^{2}m^{2}m^{1}2$	√			$^{2}\bar{1}$	
		$^{1}2/^{2}m$				$^{1}1$	
		^{2}m	√		$^{1}6/^{2}m$ ($6/m'$)	$^{1}6/^{2}m$	
		$^{1}2$				$^{2}\bar{6}$	√
		$^{2}\bar{1}$				$^{1}6$	
		$^{1}1$				$^{2}\bar{3}$	

$\mathcal{G}$	$\mathcal{R}_s$ (MPG)	$\mathcal{R}_s^{'}$	AM	$\mathcal{G}$	$\mathcal{R}_s$ (MPG)	$\mathcal{R}_s^{'}$	AM
$6/m$ ($\mathcal{C}_{6h}$)	${}^{1}6/{}^{2}m$ ($6/m'$)	${}^{1}3$		$6/mmm$ ($\mathcal{D}_{6h}$)	${}^{1}6/{}^{2}m{}^{1}m{}^{1}m$ ($6/m'mm$)	${}^{2}\bar{6}$	$\surd$
		${}^{1}2/{}^{2}m$				${}^{1}6$	
		${}^{2}m$	$\surd$			${}^{2}\bar{3}{}^{1}m$	
		${}^{1}2$				${}^{1}3{}^{1}m$	
		${}^{2}\bar{1}$				${}^{1}3{}^{2}2$	$\surd$
		${}^{1}1$				${}^{2}\bar{3}$	
$6/mmm$ ($\mathcal{D}_{6h}$)	${}^{2}6/{}^{1}m{}^{2}m{}^{1}m$ ($6'/mm'm$)	${}^{2}6/{}^{1}m{}^{2}m{}^{1}m$				${}^{1}3$	
		${}^{1}\bar{6}{}^{2}m{}^{2}2$	$\surd$			${}^{2}m{}^{1}m{}^{1}m$	
		${}^{1}\bar{6}{}^{1}m{}^{1}2$				${}^{1}m{}^{1}m{}^{1}2$	
		${}^{2}6{}^{1}m{}^{2}m$	$\surd$			${}^{2}2{}^{2}2{}^{1}2$	$\surd$
		${}^{2}6{}^{1}2{}^{2}2$	$\surd$			${}^{1}2/{}^{2}m$	
		${}^{2}6/{}^{1}m$				${}^{2}2/{}^{1}m$	
		${}^{1}\bar{6}$				${}^{2}m$	$\surd$
		${}^{2}6$	$\surd$			${}^{1}m$	
		${}^{2}\bar{3}{}^{2}m$				${}^{2}2$	$\surd$
		${}^{2}\bar{3}{}^{1}m$				${}^{1}2$	
		${}^{1}3{}^{2}m$	$\surd$			${}^{2}\bar{1}$	
		${}^{1}3{}^{1}m$				${}^{1}1$	
		${}^{1}3{}^{2}2$	$\surd$		${}^{1}6/{}^{2}m{}^{2}m{}^{2}m$ ($6/m'm'm'$)	${}^{1}6/{}^{2}m{}^{2}m{}^{2}m$	
		${}^{1}3{}^{1}2$				${}^{2}\bar{6}{}^{2}m{}^{1}2$	$\surd$
		${}^{2}\bar{3}$				${}^{1}6{}^{2}m{}^{2}m$	$\surd$
		${}^{1}3$				${}^{1}6{}^{1}2{}^{1}2$	
		${}^{2}m{}^{1}m{}^{1}m$				${}^{1}6/{}^{2}m$	
		${}^{2}m{}^{1}m{}^{2}2$	$\surd$			${}^{2}\bar{6}$	$\surd$
		${}^{2}2{}^{2}2{}^{1}2$	$\surd$			${}^{1}6$	
		${}^{1}2/{}^{2}m$				${}^{2}\bar{3}{}^{2}m$	
		${}^{2}2/{}^{1}m$				${}^{1}3{}^{2}m$	$\surd$
		${}^{2}m$	$\surd$			${}^{1}3{}^{1}2$	
		${}^{1}m$				${}^{2}\bar{3}$	
		${}^{2}2$	$\surd$			${}^{1}3$	
		${}^{1}2$				${}^{2}m{}^{2}m{}^{2}m$	
		${}^{2}\bar{1}$				${}^{2}m{}^{2}m{}^{1}2$	$\surd$
		${}^{1}1$				${}^{1}2{}^{1}2{}^{1}2$	
	${}^{1}6/{}^{2}m{}^{1}m{}^{1}m$ ($6/m'mm$)	${}^{1}6/{}^{2}m{}^{1}m{}^{1}m$				${}^{1}2/{}^{2}m$	
		${}^{2}\bar{6}/{}^{1}m{}^{2}2$	$\surd$			${}^{2}m$	$\surd$
		${}^{1}6{}^{1}m{}^{1}m$				${}^{1}2$	
		${}^{1}6{}^{2}2{}^{2}2$	$\surd$			${}^{2}\bar{1}$	
		${}^{1}6/{}^{2}m$				${}^{1}1$	

$\mathcal{G}$	$\mathcal{R}_s$ (MPG)	$\mathcal{R}_s^{'}$	AM	$\mathcal{G}$	$\mathcal{R}_s$ (MPG)	$\mathcal{R}_s^{'}$	AM
$m\bar{3}$ ($\mathcal{T}_h$)	${}^2m^2\bar{3}$ ($m'\bar{3}'$)	${}^2m^2\bar{3}$		$m\bar{3}m$ ($\mathcal{O}_h$)	${}^2m^2\bar{3}^1m$ ($m'\bar{3}'m$)	${}^22/{}^1m$	
		${}^12^13$				${}^2m$	$\surd$
		${}^2\bar{3}$				${}^1m$	
		^{13}				^{22}	$\surd$
		${}^2m^2m^2m$				^{12}	
		${}^2m^2m^12$	$\surd$			${}^2\bar{1}$	
		${}^12^12^12$				^{11}	
		${}^12/{}^2m$			${}^2m^2\bar{3}^2m$ ($m'\bar{3}'m'$)	${}^2m^2\bar{3}^2m$	
		${}^2m$	$\surd$			${}^2\bar{4}^13^2m$	$\surd$
		^{12}				${}^14^13^12$	
		${}^2\bar{1}$				${}^2m^2\bar{3}$	
		^{11}				${}^12^13$	
$m\bar{3}m$ ($\mathcal{O}_h$)	${}^2m^2\bar{3}^1m$ ($m'\bar{3}'m$)	${}^2m^2\bar{3}^1m$				${}^2\bar{3}^2m$	
		${}^1\bar{4}^13^1m$				${}^13^2m$	$\surd$
		${}^24^13^22$	$\surd$			${}^13^12$	
		${}^2m^2\bar{3}$				${}^2\bar{3}$	
		${}^12^13$				^{13}	
		${}^2\bar{3}^1m$				${}^14/{}^2m^2m^2m$	
		${}^13^1m$				${}^2\bar{4}^12^2m$	$\surd$
		${}^13^22$	$\surd$			${}^14^2m^2m$	$\surd$
		${}^2\bar{3}$				${}^14^12^12$	
		^{13}				${}^14/{}^2m$	
		${}^24/{}^2m^2m^1m$				${}^2\bar{4}$	$\surd$
		${}^1\bar{4}^22^2m$	$\surd$			^{14}	
		${}^1\bar{4}^12^1m$				${}^2m^2m^2m$	
		${}^24^2m^1m$	$\surd$			${}^2m^2m^12$	$\surd$
		${}^24^12^22$	$\surd$			${}^12^12^12$	
		${}^24/{}^2m$				${}^12/{}^2m$	
		${}^1\bar{4}$				${}^2m$	$\surd$
		^{24}	$\surd$			^{12}	
		${}^2m^2m^2m$				${}^2\bar{1}$	
		${}^2m^1m^1m$				^{11}	
		${}^2m^2m^12$	$\surd$				
		${}^1m^1m^12$					
		${}^22^22^12$	$\surd$				
		${}^12^12^12$					
		${}^12/{}^2m$					

Tab. S3: 82 potential pathways for magnetic transitions from antiferromagnetic phase to altermagnetic phase in PT-symmetric antiferromagnets. Here, $\mathcal{G}$ represents the crystallographic point group, while $\mathcal{R}_s$ denotes the spin point groups. When symmetry is broken, the resulting spin point groups are labeled as $\mathcal{R}_s^{'}$. Subsequently, the altermagnetic classification is given in the fourth column, where d, g, i denote the d-, g-, i-wave symmetries of the spin-dependent Fermi surface, respectively. The last column indicates the number of spin point groups.

$\mathcal{G}$	$\mathcal{R}_s$	$\mathcal{R}_s'$	Altermagnetism classification	number
$2/m$ ($\mathcal{C}_{2h}$)	${}^{2}2/{}^{1}m$	${}^{2}2$	d	1
	${}^{1}2/{}^{2}m$	${}^{2}m$	d	1
mmm ($\mathcal{D}_{2h}$)	${}^{1}m{}^{1}m{}^{2}m$	${}^{2}m{}^{1}m{}^{2}2$; ${}^{2}2{}^{2}2{}^{1}2$; ${}^{2}m$; ${}^{2}2$	d; d; d; d	4
	${}^{2}m{}^{2}m{}^{2}m$	${}^{2}m{}^{2}m{}^{1}2$; ${}^{2}m$	d; d	2
$4/m$ ($\mathcal{C}_{4h}$)	${}^{1}4/{}^{2}m$	${}^{2}\bar{4}$; ${}^{2}m$	d; d	2
	${}^{2}4/{}^{2}m$	${}^{2}4$; ${}^{2}m$	d; d	2
$4/mmm$ ($\mathcal{D}_{4h}$)	${}^{1}4/{}^{2}m{}^{1}m{}^{1}m$	${}^{2}\bar{4}{}^{2}2{}^{1}m$; ${}^{1}4{}^{2}2{}^{2}2$; ${}^{2}\bar{4}$; ${}^{2}m{}^{1}m{}^{2}2$; ${}^{2}2{}^{2}2{}^{1}2$; ${}^{2}m$; ${}^{2}2$	d; g; d; d; d; d; d	7
	${}^{2}4/{}^{2}m{}^{2}m{}^{1}m$	${}^{1}\bar{4}{}^{2}2{}^{2}m$; ${}^{2}4{}^{2}m{}^{1}m$; ${}^{2}4{}^{1}2{}^{2}2$; ${}^{2}4$; ${}^{2}m{}^{2}m{}^{1}2$; ${}^{2}2{}^{2}2{}^{1}2$; ${}^{2}m$; ${}^{2}2$	g; d; d; d; d; d; d; d	8
	${}^{1}4/{}^{2}m{}^{2}m{}^{2}m$	${}^{2}\bar{4}{}^{1}2{}^{2}m$; ${}^{1}4{}^{2}m{}^{2}m$; ${}^{2}\bar{4}$; ${}^{2}m{}^{2}m{}^{1}2$; ${}^{2}m$	d; g; d; d; d	5
$\bar{3}m$ ($\mathcal{D}_{3d}$)	${}^{2}\bar{3}{}^{1}m$	${}^{1}3{}^{2}2$; ${}^{2}2$	g; d	2
	${}^{2}\bar{3}{}^{2}m$	${}^{1}3{}^{2}m$; ${}^{2}m$	g; d	2
$6/m$ ($\mathcal{C}_{6h}$)	${}^{2}6/{}^{1}m$	${}^{2}6$; ${}^{2}2$	g; d	2
	${}^{1}6/{}^{2}m$	${}^{2}\bar{6}$; ${}^{2}m$	g; d	2
$6/mmm$ ($\mathcal{D}_{6h}$)	${}^{1}6/{}^{2}m{}^{1}m{}^{1}m$	${}^{2}\bar{6}{}^{1}m{}^{2}2$; ${}^{1}6{}^{2}2{}^{2}2$; ${}^{2}\bar{6}$; ${}^{1}3{}^{2}2$; ${}^{2}2{}^{2}2{}^{1}2$; ${}^{2}m$; ${}^{2}2$	g; i; g; g; d; d; d	7
	${}^{2}6/{}^{1}m{}^{2}m{}^{1}m$	${}^{1}\bar{6}{}^{2}m{}^{2}2$; ${}^{2}6{}^{1}m{}^{2}m$; ${}^{2}6{}^{1}2{}^{2}2$; ${}^{2}6$; ${}^{1}3{}^{2}m$; ${}^{1}3{}^{2}2$; ${}^{2}m{}^{1}m{}^{2}2$; ${}^{2}2{}^{2}2{}^{1}2$; ${}^{2}m$; ${}^{2}2$	i; g; g; g; g; g; g; d; d; d; d	10
	${}^{1}6/{}^{2}m{}^{2}m{}^{2}m$	${}^{2}\bar{6}{}^{2}m{}^{1}2$; ${}^{1}6{}^{2}m{}^{2}m$; ${}^{2}\bar{6}$; ${}^{1}3{}^{2}m$; ${}^{2}m{}^{2}m{}^{1}2$; ${}^{2}m$	g; i; g; g; d; d	6
$m\bar{3}$ ($\mathcal{T}_h$)	${}^{2}m{}^{2}\bar{3}$	${}^{2}m{}^{2}m{}^{1}2$; ${}^{2}m$	d; d	2
$m\bar{3}m$ ($\mathcal{O}_h$)	${}^{2}m{}^{2}\bar{3}{}^{1}m$	${}^{2}4{}^{1}3{}^{2}2$; ${}^{1}3{}^{2}2$; ${}^{1}\bar{4}{}^{2}2{}^{2}m$; ${}^{2}4{}^{2}m{}^{1}m$; ${}^{2}4$; ${}^{2}4{}^{1}2{}^{2}2$; ${}^{2}m{}^{2}m{}^{1}2$; ${}^{2}2{}^{2}2{}^{1}2$; ${}^{2}m$; ${}^{2}2$	i; g; g; d; d; d; d; d; d; d	10
	${}^{2}m{}^{2}\bar{3}{}^{2}m$	${}^{2}\bar{4}{}^{1}3{}^{2}m$; ${}^{1}3{}^{2}m$; ${}^{2}\bar{4}{}^{1}2{}^{2}m$; ${}^{1}4{}^{2}m{}^{2}m$; ${}^{2}\bar{4}$; ${}^{2}m{}^{2}m{}^{1}2$; ${}^{2}m$	i; g; d; g; d; d; d	7

Tab. S4: A total of 318 collinear spin space groups (155 groups without $[C_2\|P]$ and 163 groups with $[C_2\|P]$) can transition from $\boldsymbol{t}T$-symmetric antiferromagnetism to altermagnetism through the breaking of specific translational symmetries. Here, $\mathcal{G}$ denotes the parent crystallographic point group. The columns labeled "without $[C_2\|P]$" and "with $[C_2\|P]$" indicate whether the spin space group lacks or preserves the $[C_2\|P]$ symmetry, respectively. The final column lists the number of spin space groups corresponding to each crystallographic point group.

$\mathcal{G}$	without $[C_2\|P]$	with $[C_2\|P]$	number
2	$P_b^1 2_1\, {}^{\infty m}1$; $P_C^1 2_1\, {}^{\infty m}1$	—	2
m	$P_c^1 c\, {}^{\infty m}1$; $P_A^1 c\, {}^{\infty m}1$; $C_c^1 c\, {}^{\infty m}1$	—	3
$2/m$	—	$P_b^1 2_1/{}^1 m\, {}^{\infty m}1$; $P_C^1 2_1/{}^1 m\, {}^{\infty m}1$; $P_c^1 2/{}^1 c\, {}^{\infty m}1$; $P_A^1 2/{}^1 c\, {}^{\infty m}1$; $P_b^1 2_1/{}^1 c\, {}^{\infty m}1$; $P_c^1 2_1/{}^1 c\, {}^{\infty m}1$; $P_A^1 2_1/{}^1 c\, {}^{\infty m}1$; $P_C^1 2_1/{}^1 c\, {}^{\infty m}1$; $C_c^1 2/{}^1 c\, {}^{\infty m}1$	9
222	$P_c^1 2^1 2^1 2_1\, {}^{\infty m}1$; $P_B^1 2^1 2^1 2_1\, {}^{\infty m}1$; $P_I^1 2^1 2^1 2_1\, {}^{\infty m}1$; $P_b^1 2_1^1 2_1^1 2\, {}^{\infty m}1$; $P_B^1 2_1^1 2_1^1 2\, {}^{\infty m}1$; $P_C^1 2_1^1 2_1^1 2\, {}^{\infty m}1$; $P_I^1 2_1^1 2_1^1 2\, {}^{\infty m}1$; $P_c^1 2_1^1 2_1^1 2_1\, {}^{\infty m}1$; $P_C^1 2_1^1 2_1^1 2_1\, {}^{\infty m}1$; $P_I^1 2_1^1 2_1^1 2_1\, {}^{\infty m}1$; $C_c^1 2^1 2^1 2_1\, {}^{\infty m}1$; $C_A^1 2^1 2^1 2_1\, {}^{\infty m}1$; $I_c^1 2^1 2^1 2\, {}^{\infty m}1$;	—	13
$mm2$	$P_c^1 m^1 c^1 2_1\, {}^{\infty m}1$; $P_A^1 m^1 c^1 2_1\, {}^{\infty m}1$; $P_B^1 m^1 c^1 2_1\, {}^{\infty m}1$; $P_I^1 m^1 c^1 2_1\, {}^{\infty m}1$; $P_c^1 c^1 c^1 2\, {}^{\infty m}1$; $P_A^1 c^1 c^1 2\, {}^{\infty m}1$; $P_I^1 c^1 c^1 2\, {}^{\infty m}1$; $P_a^1 m^1 a^1 2\, {}^{\infty m}1$; $P_B^1 m^1 a^1 2\, {}^{\infty m}1$; $P_C^1 m^1 a^1 2\, {}^{\infty m}1$; $P_I^1 m^1 a^1 2\, {}^{\infty m}1$; $P_a^1 c^1 a^1 2_1\, {}^{\infty m}1$; $P_c^1 c^1 a^1 2_1\, {}^{\infty m}1$; $P_A^1 c^1 a^1 2_1\, {}^{\infty m}1$; $P_B^1 c^1 a^1 2_1\, {}^{\infty m}1$; $P_C^1 c^1 a^1 2_1\, {}^{\infty m}1$; $P_I^1 c^1 a^1 2_1\, {}^{\infty m}1$; $P_b^1 n^1 c^1 2\, {}^{\infty m}1$; $P_c^1 n^1 c^1 2\, {}^{\infty m}1$; $P_A^1 n^1 c^1 2\, {}^{\infty m}1$; $P_B^1 n^1 c^1 2\, {}^{\infty m}1$; $P_C^1 n^1 c^1 2\, {}^{\infty m}1$; $P_I^1 n^1 c^1 2\, {}^{\infty m}1$; $P_a^1 m^1 n^1 2_1\, {}^{\infty m}1$; $P_c^1 m^1 n^1 2_1\, {}^{\infty m}1$; $P_A^1 m^1 n^1 2_1\, {}^{\infty m}1$; $P_B^1 m^1 n^1 2_1\, {}^{\infty m}1$; $P_C^1 m^1 n^1 2_1\, {}^{\infty m}1$; $P_I^1 m^1 n^1 2_1\, {}^{\infty m}1$; $P_b^1 b^1 a^1 2\, {}^{\infty m}1$; $P_C^1 b^1 a^1 2\, {}^{\infty m}1$; $P_A^1 b^1 a^1 2\, {}^{\infty m}1$; $P_I^1 b^1 a^1 2\, {}^{\infty m}1$; $P_a^1 n^1 a^1 2_1\, {}^{\infty m}1$; $P_b^1 n^1 a^1 2_1\, {}^{\infty m}1$; $P_c^1 n^1 a^1 2_1\, {}^{\infty m}1$; $P_A^1 n^1 a^1 2_1\, {}^{\infty m}1$; $P_B^1 n^1 a^1 2_1\, {}^{\infty m}1$; $P_C^1 n^1 a^1 2_1\, {}^{\infty m}1$; $P_I^1 n^1 a^1 2_1\, {}^{\infty m}1$; $P_a^1 n^1 n^1 2\, {}^{\infty m}1$; $P_c^1 n^1 n^1 2\, {}^{\infty m}1$; $P_A^1 n^1 n^1 2\, {}^{\infty m}1$; $P_C^1 n^1 n^1 2\, {}^{\infty m}1$; $P_I^1 n^1 n^1 2\, {}^{\infty m}1$; $C_c^1 m^1 c^1 2_1\, {}^{\infty m}1$; $C_A^1 m^1 c^1 2_1\, {}^{\infty m}1$; $C_c^1 c^1 c^1 2\, {}^{\infty m}1$; $C_A^1 c^1 c^1 2\, {}^{\infty m}1$; $A_b^1 e^1 m^1 2\, {}^{\infty m}1$; $A_B^1 e^1 m^1 2\, {}^{\infty m}1$; $A_a^1 m^1 a^1 2\, {}^{\infty m}1$; $A_B^1 m^1 a^1 2\, {}^{\infty m}1$; $A_a^1 e^1 a^1 2\, {}^{\infty m}1$; $A_b^1 e^1 a^1 2\, {}^{\infty m}1$; $A_B^1 e^1 a^1 2\, {}^{\infty m}1$; $I_c^1 b^1 a^1 2\, {}^{\infty m}1$; $I_a^1 b^1 a^1 2\, {}^{\infty m}1$; $I_c^1 m^1 a^1 2\, {}^{\infty m}1$; $I_a^1 m^1 a^1 2\, {}^{\infty m}1$; $I_b^1 m^1 a^1 2\, {}^{\infty m}1$	—	61
mmm	—	$P_c^1 n^1 n^1 n\, {}^{\infty m}1$; $P_C^1 n^1 n^1 n\, {}^{\infty m}1$; $P_I^1 n^1 n^1 n\, {}^{\infty m}1$; $P_c^1 c^1 c^1 m\, {}^{\infty m}1$; $P_B^1 c^1 c^1 m\, {}^{\infty m}1$; $P_I^1 c^1 c^1 m\, {}^{\infty m}1$; $P_a^1 b^1 a^1 n\, {}^{\infty m}1$; $P_A^1 b^1 a^1 n\, {}^{\infty m}1$; $P_C^1 b^1 a^1 n\, {}^{\infty m}1$; $P_I^1 b^1 a^1 n\, {}^{\infty m}1$; $P_a^1 m^1 m^1 a\, {}^{\infty m}1$; $P_B^1 m^1 m^1 a\, {}^{\infty m}1$; $P_C^1 m^1 m^1 a\, {}^{\infty m}1$; $P_I^1 m^1 m^1 a\, {}^{\infty m}1$; $P_a^1 n^1 n^1 a\, {}^{\infty m}1$; $P_b^1 n^1 n^1 a\, {}^{\infty m}1$; $P_c^1 n^1 n^1 a\, {}^{\infty m}1$; $P_A^1 n^1 n^1 a\, {}^{\infty m}1$; $P_B^1 n^1 n^1 a\, {}^{\infty m}1$; $P_C^1 n^1 n^1 a\, {}^{\infty m}1$; $P_I^1 n^1 n^1 a\, {}^{\infty m}1$; $P_a^1 m^1 n^1 a\, {}^{\infty m}1$; $P_c^1 m^1 n^1 a\, {}^{\infty m}1$; $P_A^1 m^1 n^1 a\, {}^{\infty m}1$; $P_B^1 m^1 n^1 a\, {}^{\infty m}1$; $P_C^1 m^1 n^1 a\, {}^{\infty m}1$; $P_I^1 m^1 n^1 a\, {}^{\infty m}1$; $P_a^1 c^1 c^1 a\, {}^{\infty m}1$; $P_c^1 c^1 c^1 a\, {}^{\infty m}1$; $P_A^1 c^1 c^1 a\, {}^{\infty m}1$; $P_B^1 c^1 c^1 a\, {}^{\infty m}1$; $P_C^1 c^1 c^1 a\, {}^{\infty m}1$; $P_I^1 c^1 c^1 a\, {}^{\infty m}1$; $P_a^1 b^1 a^1 m\, {}^{\infty m}1$; $P_A^1 b^1 a^1 m\, {}^{\infty m}1$; $P_C^1 b^1 a^1 m\, {}^{\infty m}1$; $P_I^1 b^1 a^1 m\, {}^{\infty m}1$; $P_b^1 c^1 c^1 n\, {}^{\infty m}1$; $P_c^1 c^1 c^1 n\, {}^{\infty m}1$; $P_A^1 c^1 c^1 n\, {}^{\infty m}1$; $P_C^1 c^1 c^1 n\, {}^{\infty m}1$; $P_I^1 c^1 c^1 n\, {}^{\infty m}1$; $P_b^1 b^1 c^1 m\, {}^{\infty m}1$; $P_c^1 b^1 c^1 m\, {}^{\infty m}1$; $P_A^1 b^1 c^1 m\, {}^{\infty m}1$; $P_B^1 b^1 c^1 m\, {}^{\infty m}1$; $P_C^1 b^1 c^1 m\, {}^{\infty m}1$; $P_I^1 b^1 c^1 m\, {}^{\infty m}1$; $P_a^1 n^1 n^1 m\, {}^{\infty m}1$; $P_c^1 n^1 n^1 m\, {}^{\infty m}1$; $P_B^1 n^1 n^1 m\, {}^{\infty m}1$; $P_C^1 n^1 n^1 m\, {}^{\infty m}1$; $P_I^1 n^1 n^1 m\, {}^{\infty m}1$; $P_b^1 m^1 m^1 n\, {}^{\infty m}1$; $P_B^1 m^1 m^1 n\, {}^{\infty m}1$; $P_C^1 m^1 m^1 n\, {}^{\infty m}1$; $P_I^1 m^1 m^1 n\, {}^{\infty m}1$; $P_a^1 b^1 c^1 n\, {}^{\infty m}1$; $P_b^1 b^1 c^1 n\, {}^{\infty m}1$; $P_c^1 b^1 c^1 n\, {}^{\infty m}1$; $P_A^1 b^1 c^1 n\, {}^{\infty m}1$; $P_B^1 b^1 c^1 n\, {}^{\infty m}1$; $P_C^1 b^1 c^1 n\, {}^{\infty m}1$; $P_I^1 b^1 c^1 n\, {}^{\infty m}1$; $P_a^1 b^1 c^1 a\, {}^{\infty m}1$; $P_C^1 b^1 c^1 a\, {}^{\infty m}1$	90

$\mathcal{G}$	without $[C_2\|\|P]$	with $[C_2\|\|P]$	number
mmm	—	$P_I^1 b^1 c^1 a\,^{\infty m}1$; $P_a^1 n^1 m^1 a\,^{\infty m}1$; $P_b^1 n^1 m^1 a\,^{\infty m}1$; $P_c^1 n^1 m^1 a\,^{\infty m}1$; $P_A^1 n^1 m^1 a\,^{\infty m}1$; $P_B^1 n^1 m^1 a\,^{\infty m}1$; $P_C^1 n^1 m^1 a\,^{\infty m}1$; $P_I^1 n^1 m^1 a\,^{\infty m}1$; $C_c^1 m^1 c^1 m\,^{\infty m}1$; $C_A^1 m^1 c^1 m\,^{\infty m}1$; $C_c^1 m^1 c^1 e\,^{\infty m}1$; $C_a^1 m^1 c^1 e\,^{\infty m}1$; $C_A^1 m^1 c^1 e\,^{\infty m}1$; $C_c^1 c^1 c^1 m\,^{\infty m}1$; $C_A^1 c^1 c^1 m\,^{\infty m}1$; $C_a^1 m^1 m^1 e\,^{\infty m}1$; $C_c^1 c^1 c^1 e\,^{\infty m}1$; $C_a^1 c^1 c^1 e\,^{\infty m}1$; $C_A^1 c^1 c^1 e\,^{\infty m}1$; $I_c^1 b^1 a^1 m\,^{\infty m}1$; $I_b^1 b^1 a^1 m\,^{\infty m}1$; $I_c^1 b^1 c^1 a\,^{\infty m}1$; $I_c^1 m^1 m^1 a\,^{\infty m}1$; $I_b^1 m^1 m^1 a\,^{\infty m}1$	
4	$P_c^1 4_1\,^{\infty m}1$; $P_I^1 4_1\,^{\infty m}1$; $P_c^1 4_2\,^{\infty m}1$; $P_I^1 4_2\,^{\infty m}1$; $P_c^1 4_3\,^{\infty m}1$; $P_I^1 4_3\,^{\infty m}1$	—	6
$4/m$	—	$P_c^1 4_2/^1 m\,^{\infty m}1$; $P_I^1 4_2/^1 m\,^{\infty m}1$; $P_C^1 4/^1 n\,^{\infty m}1$; $P_I^1 4/^1 n\,^{\infty m}1$; $P_c^1 4_2/^1 n\,^{\infty m}1$; $P_C^1 4_2/^1 n\,^{\infty m}1$; $P_I^1 4_2/^1 n\,^{\infty m}1$; $I_c^1 4_1/^1 a\,^{\infty m}1$	8
422	$P_C^1 4^1 2_1^1 2\,^{\infty m}1$; $P_I^1 4^1 2_1^1 2\,^{\infty m}1$; $P_c^1 4_1^1 2^1 2\,^{\infty m}1$; $P_I^1 4_1^1 2^1 2\,^{\infty m}1$; $P_c^1 4_1^1 2_1^1 2\,^{\infty m}1$; $P_C^1 4_1^1 2_1^1 2\,^{\infty m}1$; $P_I^1 4_1^1 2_1^1 2\,^{\infty m}1$; $P_c^1 4_2^1 2^1 2\,^{\infty m}1$; $P_I^1 4_2^1 2^1 2\,^{\infty m}1$; $P_c^1 4_2^1 2_1^1 2\,^{\infty m}1$; $P_C^1 4_2^1 2_1^1 2\,^{\infty m}1$; $P_I^1 4_2^1 2_1^1 2\,^{\infty m}1$; $P_c^1 4_3^1 2^1 2\,^{\infty m}1$; $P_I^1 4_3^1 2^1 2\,^{\infty m}1$; $P_c^1 4_3^1 2_1^1 2\,^{\infty m}1$; $P_C^1 4_3^1 2_1^1 2\,^{\infty m}1$; $P_I^1 4_3^1 2_1^1 2\,^{\infty m}1$	—	17
$4mm$	$P_C^1 4^1 b^1 m\,^{\infty m}1$; $P_I^1 4^1 b^1 m\,^{\infty m}1$; $P_c^1 4_2^1 c^1 m\,^{\infty m}1$; $P_I^1 4_2^1 c^1 m\,^{\infty m}1$; $P_c^1 4_2^1 n^1 m\,^{\infty m}1$; $P_C^1 4_2^1 n^1 m\,^{\infty m}1$; $P_I^1 4_2^1 n^1 m\,^{\infty m}1$; $P_c^1 4^1 c^1 c\,^{\infty m}1$; $P_I^1 4^1 c^1 c\,^{\infty m}1$; $P_c^1 4^1 n^1 c\,^{\infty m}1$; $P_C^1 4^1 n^1 c\,^{\infty m}1$; $P_I^1 4^1 n^1 c\,^{\infty m}1$; $P_c^1 4_2^1 m^1 c\,^{\infty m}1$; $P_I^1 4_2^1 m^1 c\,^{\infty m}1$; $P_c^1 4_2^1 b^1 c\,^{\infty m}1$; $P_C^1 4_2^1 b^1 c\,^{\infty m}1$; $P_I^1 4_2^1 b^1 c\,^{\infty m}1$; $I_c^1 4^1 m^1 m\,^{\infty m}1$; $I_c^1 4_1^1 c^1 d\,^{\infty m}1$	—	19
$\bar{4}2m$	$P_c^1 \bar{4}^1 2^1 c\,^{\infty m}1$; $P_I^1 \bar{4}^1 2^1 c\,^{\infty m}1$; $P_C^1 \bar{4}^1 2_1^1 m\,^{\infty m}1$; $P_I^1 \bar{4}^1 2_1^1 m\,^{\infty m}1$; $P_c^1 \bar{4}^1 2_1^1 c\,^{\infty m}1$; $P_C^1 \bar{4}^1 2_1^1 c\,^{\infty m}1$; $P_I^1 \bar{4}^1 2_1^1 c\,^{\infty m}1$; $P_c^1 \bar{4}^1 c^1 2\,^{\infty m}1$; $P_I^1 \bar{4}^1 c^1 2\,^{\infty m}1$; $P_C^1 \bar{4}^1 b^1 2\,^{\infty m}1$; $P_I^1 \bar{4}^1 b^1 2\,^{\infty m}1$; $P_c^1 \bar{4}^1 n^1 2\,^{\infty m}1$; $P_C^1 \bar{4}^1 n^1 2\,^{\infty m}1$; $P_I^1 \bar{4}^1 n^1 2\,^{\infty m}1$; $I_c^1 \bar{4}^1 c^1 2\,^{\infty m}1$	—	15
$4/mmm$	—	$P_c^1 4/^1 m^1 c^1 c\,^{\infty m}1$; $P_I^1 4/^1 m^1 c^1 c\,^{\infty m}1$; $P_C^1 4/^1 n^1 b^1 m\,^{\infty m}1$; $P_I^1 4/^1 n^1 b^1 m\,^{\infty m}1$; $P_c^1 4/^1 n^1 n^1 c\,^{\infty m}1$; $P_C^1 4/^1 n^1 n^1 c\,^{\infty m}1$; $P_I^1 4/^1 n^1 n^1 c\,^{\infty m}1$; $P_C^1 4/^1 m^1 b^1 m\,^{\infty m}1$; $P_I^1 4/^1 m^1 b^1 m\,^{\infty m}1$; $P_c^1 4/^1 m^1 n^1 c\,^{\infty m}1$; $P_C^1 4/^1 m^1 n^1 c\,^{\infty m}1$; $P_I^1 4/^1 m^1 n^1 c\,^{\infty m}1$; $P_C^1 4/^1 n^1 m^1 m\,^{\infty m}1$; $P_I^1 4/^1 n^1 m^1 m\,^{\infty m}1$; $P_c^1 4/^1 n^1 c^1 c\,^{\infty m}1$; $P_C^1 4/^1 n^1 c^1 c\,^{\infty m}1$; $P_I^1 4/^1 n^1 c^1 c\,^{\infty m}1$; $P_c^1 4_2/^1 m^1 m^1 c\,^{\infty m}1$; $P_I^1 4_2/^1 m^1 m^1 c\,^{\infty m}1$; $P_c^1 4_2/^1 m^1 c^1 m\,^{\infty m}1$; $P_I^1 4_2/^1 m^1 c^1 m\,^{\infty m}1$; $P_c^1 4_2/^1 n^1 b^1 c\,^{\infty m}1$; $P_C^1 4_2/^1 n^1 b^1 c\,^{\infty m}1$; $P_I^1 4_2/^1 n^1 b^1 c\,^{\infty m}1$; $P_c^1 4_2/^1 n^1 n^1 m\,^{\infty m}1$; $P_C^1 4_2/^1 n^1 n^1 m\,^{\infty m}1$; $P_I^1 4_2/^1 n^1 n^1 m\,^{\infty m}1$; $P_c^1 4_2/^1 m^1 b^1 c\,^{\infty m}1$; $P_C^1 4_2/^1 m^1 b^1 c\,^{\infty m}1$; $P_I^1 4_2/^1 m^1 b^1 c\,^{\infty m}1$; $P_c^1 4_2/^1 m^1 n^1 m\,^{\infty m}1$; $P_C^1 4_2/^1 m^1 n^1 m\,^{\infty m}1$; $P_I^1 4_2/^1 m^1 n^1 m\,^{\infty m}1$; $P_c^1 4_2/^1 n^1 m^1 c\,^{\infty m}1$; $P_C^1 4_2/^1 n^1 m^1 c\,^{\infty m}1$; $P_I^1 4_2/^1 n^1 m^1 c\,^{\infty m}1$; $P_c^1 4_2/^1 n^1 c^1 m\,^{\infty m}1$; $P_C^1 4_2/^1 n^1 c^1 m\,^{\infty m}1$; $P_I^1 4_2/^1 n^1 c^1 m\,^{\infty m}1$; $I_c^1 4/^1 m^1 c^1 m\,^{\infty m}1$; $I_c^1 4_1/^1 a^1 m^1 d\,^{\infty m}1$; $I_c^1 4_1/^1 a^1 c^1 d\,^{\infty m}1$	42
$3m$	$P_c^1 3^1 c^1 1\,^{\infty m}1$; $P_c^1 3^1 1^1 c\,^{\infty m}1$; $R_I^1 3^1 c\,^{\infty m}1$	—	3
$\bar{3}m$	—	$P_c^1 \bar{3}^1 1^1 c\,^{\infty m}1$; $P_c^1 \bar{3}^1 c^1 1\,^{\infty m}1$; $R_c^1 \bar{3}^1 c\,^{\infty m}1$;	3
6	$P_c^1 6_1\,^{\infty m}1$; $P_c^1 6_5\,^{\infty m}1$; $P_c^1 6_3\,^{\infty m}1$	—	3
$6/m$	—	$P_c^1 6_3/^1 m\,^{\infty m}1$	1

$\mathcal{G}$	without $[C_2\|P]$	with $[C_2\|P]$	number
622	$P_c^1 6_1^1 2^1 2\,{}^{\infty m}1$; $P_c^1 6_5^1 2^1 2\,{}^{\infty m}1$; $P_c^1 6_3^1 2^1 2\,{}^{\infty m}1$	—	3
$6mm$	$P_c^1 6^1 c^1 c\,{}^{\infty m}1$; $P_c^1 6_3^1 c^1 m\,{}^{\infty m}1$; $P_c^1 6_3^1 m^1 c\,{}^{\infty m}1$	—	3
$\bar{6}m2$	$P_c^1 \bar{6}^1 c^1 2\,{}^{\infty m}1$; $P_c^1 \bar{6}^1 2^1 c\,{}^{\infty m}1$	—	2
$6/mmm$	—	$P_c^1 6/{}^1 m^1 c^1 c\,{}^{\infty m}1$; $P_c^1 6_3/{}^1 m^1 c^1 m\,{}^{\infty m}1$; $P_c^1 6_3/{}^1 m^1 m^1 c\,{}^{\infty m}1$	3
23	$P_I\,{}^1 2_1\,{}^1 3\,{}^{\infty m}1$	—	1
$m\bar{3}$	—	$P_I^1 n^1 \bar{3}\,{}^{\infty m}1$; $P_I^1 a^1 \bar{3}\,{}^{\infty m}1$	2
$\bar{4}3m$	$P_I^1 \bar{4}^1 3^1 n\,{}^{\infty m}1$	—	1
432	$P_I^1 4_2^1 3^1 2\,{}^{\infty m}1$; $P_I^1 4_3^1 3^1 2\,{}^{\infty m}1$; $P_I^1 4_1^1 3^1 2\,{}^{\infty m}1$	—	3
$m\bar{3}m$	—	$P_I^1 n^1 \bar{3}^1 n\,{}^{\infty m}1$; $P_I^1 m^1 \bar{3}^1 n\,{}^{\infty m}1$; $P_I^1 n^1 \bar{3}^1 m\,{}^{\infty m}1$; $F_S^1 m^1 \bar{3}^1 c\,{}^{\infty m}1$; $F_S^1 d^1 \bar{3}^1 c\,{}^{\infty m}1$	5

II. Electric field manipulation of 2D materials

Electric field manipulation is an efficient method to drive the transition from antiferromagnetism to altermagnetism in 2D materials. Here, we demonstrate this using the 2D antiferromagnetic MoTe and its analogues. It is important to note that some previous studies[2, 3] have proposed that an external electric field can induce spin splitting in 2D systems. However, the symmetry-breaking mechanisms in these works differ fundamentally from those in the magnetic transition from antiferromagnetism to altermagnetism. In these studies, magnetic atoms with opposite spins are arranged in distinct planes along the direction of the applied electric field. When the electric field is applied, the electric potentials of the opposite spin magnetic sublattices become unequal, breaking any symmetry operations that would normally connect these sublattices. This lack of symmetry between the opposite spin sublattices typically leads to energy band splitting across the entire Brillouin zone. In contrast, to achieve a magnetic transition from antiferromagnetism to altermagnetism through an applied electric field, the opposite spin magnetic atoms must lie in the same plane. Furthermore, at least one symmetry operation, denoted as $[C_2\|A]$ (where A is not space inversion and C_2 is the spin reversal operation in spin space), is required to relate the opposite spin magnetic atoms. This symmetry results in an altermagnetic system with alternating spin splitting. Additionally, these symmetry operations enforce energy degeneracy between opposite spins at the invariant high-symmetry paths.

A. Magnetic transition induced by vertical electric field for 2D MoTe

Here we take a 2D material of MoTe as an example to concretely discuss the magnetic transition from antiferromagnets to altermagnets, and other examples can be found in Fig. S19. The newly discovered monolayer MoTe shares the same crystal structure as the experimentally synthesized 2D superconductor FeSe. The monolayer consists of three layers of atoms arranged as Te-Mo-Te with a square lattice structure, as shown in Figs. S11(a)-S11(c). The energies of the monolayer with different magnetic orders of Mo atoms (details can be found in Fig. S15 and

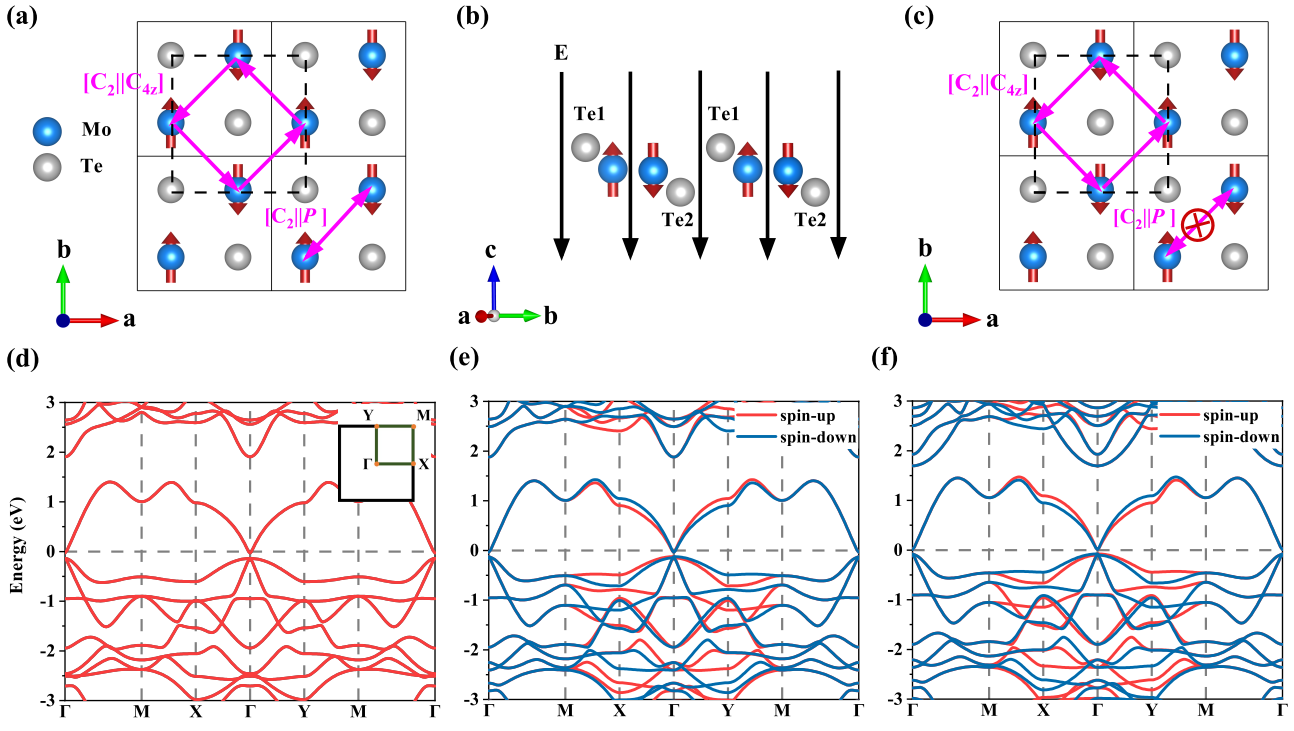


Fig. S11: The crystal structure and the energy band structure of MoTe. (a) Top view of 2D MoTe, in which the opposite spin sublattices are connected by both C_{4z} and inversion P. (b) and (c) Side view and top view of MoTe with an vertical electric field. The electric field break the inversion P. (d)-(f) Energy band structure with an vertical electric field of 0, 0.3 and -0.3 V/Å, respectively

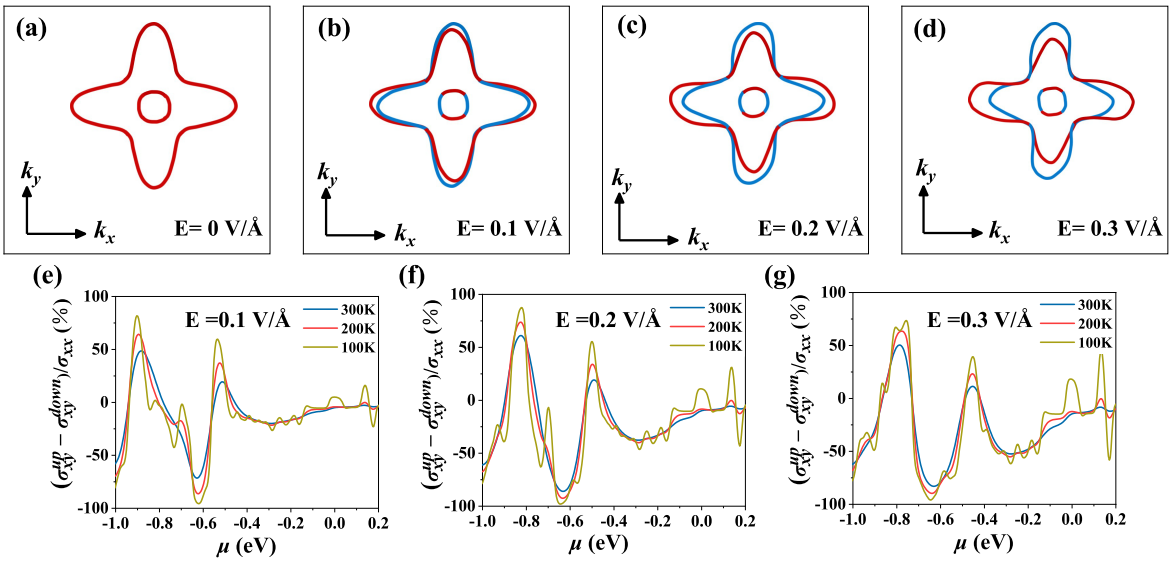


Fig. S12: (a)-(d) Energy contours of the energy bands of MoTe at a chemical potential of μ = -0.4 eV under various electric fields. (e)-(g) Charge-spin conversion ratio of MoTe at different temperatures, with electric fields of 0.1, 0.2, and 0.3 V/Å, respectively.

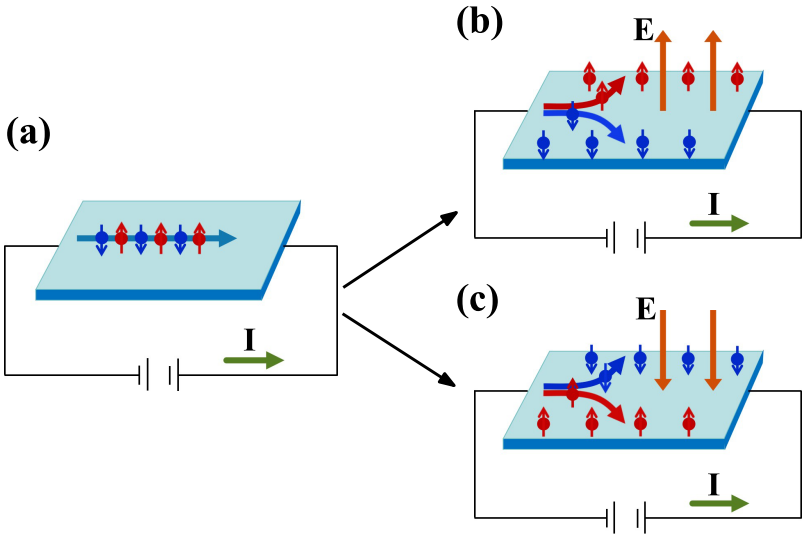


Fig. S13: Schematic of spin current generation via a vertical electric field based on 2D MoTe. (a) A charge current is generated with equal numbers of spin-up and spin-down electrons when a longitudinal electric field is applied. (b) and (c) With an additional vertical electric field, electrons with opposite spins deflect in opposite directions. The direction of the electric field can alter the deflection direction for electrons of a single spin.

Tab. S5) indicate that the antiferromagnetic coupling of nearest neighbor Mo atoms with local magnetic moment of 4.0 μ_B is the magnetic ground state. The first-principles phonon spectrum and molecular dynamics simulations demonstrate that the material is dynamically and thermodynamically stable (see section of computational details and Figs. S16 and S17 for stability information). The magnetic crystal symmetries of the ground state is the third type of nontrivial spin point group ${}^{2}4/{}^{1}m^{2}m^{1}m$. There are two prominent symmetry operations connecting the opposite spin sublattices: C_{4z} and inversion P, as shown in Fig. S11(a). Given that the coordinated layers of Te atoms share the same symmetries as the magnetic atoms, the monolayer inherently qualifies as a typical antiferromagnetic material with time-reversal symmetry. The energy band structure as shown in Fig. S11(d) reveals that the opposite spins are degenerate enforced by the PT compound operation. Upon introducing a vertical electric field, as all magnetic atoms are located within the same plane along z axis, their electric potentials are equal. However, Te1 and Te2

layers as shown in Fig. S11(b) will exhibit different electric potentials due to their distinct z coordinates, resulting in different electronic crystal potentials for opposite spin magnetic Mo atoms. Consequently, the inversion operation is broken, leaving only the C_4 rotation operation to connect the opposite spin sublattice, as shown in Fig. S11(c). Upon scrutinizing all symmetry operations of the magnets, we conclude that they can only be described by the nontrivial spin group ${}^{2}4^{2}m^{1}m$ with the breaking of PT symmetry. According to the general symmetry breaking theory, the material transform from antiferromagnetism to altermagnetism, as shown in Fig. S11. The energy band structures with vertical electric fields of +0.3 and -0.3 V/$\mathring{A}$ are illustrated in Figs. S11(e) and S11(f). The energy bands show evident characteristics of altermagnetism with alternating splitting of opposite spins. Furthermore, it is revealed that the spin-splitting can be reversed by applying opposite vertical electric field, which will be feasible for control the spin current in spintronics devices by purely electric means. It is worth noting that the spin-orbit coupling (SOC) does not strongly influence the spin splitting in the material, as shown in Fig. S18.

The well-established feature of altermagnetic materials is their spin-momentum locking that can be harnessed to induce spin currents under an electric field [4–10]. However, the mechanism for generating spin current in altermagnetic materials differs from the conventional spin Hall effect that relativistic SOC plays a pivotal role in producing transverse spin currents. In altermagnetic materials, the spin current only requires non-relativistic spin-polarized electronic states, a phenomenon known as the spin-splitter effect[11], primarily associated with opposite-spin electronic states at the Fermi level. Here, we examine how the Fermi surface splitting between opposite spins evolves with increasing vertical electrc fields for MoTe. The results shown in Figs. S12(a)-S12(d) indicate that the vertical electric field breaks the degenerate of opposite spins on the Fermi surface. Moreover, along the increase in the electric field strength, the splitting between the up and down spins becomes increasingly pronounced. The results also indicate that the increase in spin splitting does not destroy the spin-momentum locking protected by the spin-group operation ($[C_2||\ C_{4z}]$). To investigate the external-electric-field-dependent spin current, the transverse spin current was obtained by combining the first-principles calculations and semiclassical Boltzmann theory[12, 13]. The charge-spin conversion ratio is expressed as: $\sigma_{xy}^{\mathrm{up}} - \sigma_{xy}^{\mathrm{down}}/\sigma_{xx}$, where x is the direction of longitudinal current. The charge-spin conversion ratios of MoTe monolayer with different vertical electric field, temperature, and chemical potential are shown in Figs. S12(e)-S12(g). It is revealed that the charge-spin conversion can be effectively adjusted by external electric field. Moreover, the highest charge-spin conversion ratios are substantially greater than those observed in collinear antiferromagnetic RuO_2[11] and the commonly studied spin Hall material Pt[14]. The electric field modulated spin-polarization along with the transition provides new concept for modern design of spintronics devices, as shown in Fig. S13. This device design differs from previous approaches that rely on the SOC effect[15, 16]. Instead, it operates independently of SOC and primarily utilizes the nonrelativistic electronic properties of materials. The electric field not only effectively control the spin degrees of freedom, but also efficiently modulate the strength of the spin-polarization. Furthermore, the symmetry breaking theory for the transition from antiferromagnetism to altermagnetism will effectively expand the researching scope for antiferromagnetic spintronics[17].

B. Computational details

Calculations in this work were performed using the Vienna ab initio simulation package (VASP)[18, 19], which is based on density functional theory (DFT). The Perdew-Burke-Ernzerhof functional at the level of generalized gradient approximation (GGA)[20] is used to deal with the exchange-correlation interactions. Considering the strong correlation effect of transition metal, the GGA + U method[21] is used for the calculations, in which the effective U value is determined by the linear response method (see Sec. IV for details). It can be found that the U_{eff} value of the d orbitals of transition metal Mo in MoTe is 3.7 eV. The plane-wave basis is used with a cutoff energy of 700 eV. The Brillouin zone is sampled with Γ-centered k mesh of size $13 \times 13 \times 1$. The crystal structure is fully optimized until the force applied on each atom is less than 0.01 eV/Å. The energy convergence criterion is set to be 10^{-8} eV. A vacuum layer of 15 Å is used to avoid interactions between neighboring layers. Ab initio molecular dynamics simulations were performed in the NVT ensemble with a time step of 20000 fs. In addition, we used the Wannier90 software package[22] to generate the maximum local Wannier function and calculate the Fermi surface of the material. Finally, the charge-spin conversion ratio of the material was calculated using the vaspkit software package[13].

C. The calculation of U_{eff} for the d orbitals of MoTe

The U_{eff} value was determined using the linear response method, with the results illustrated in Fig. S14. The calculated effective U value is 3.742 eV, which was therefore chosen for the d orbitals of Mo atoms.

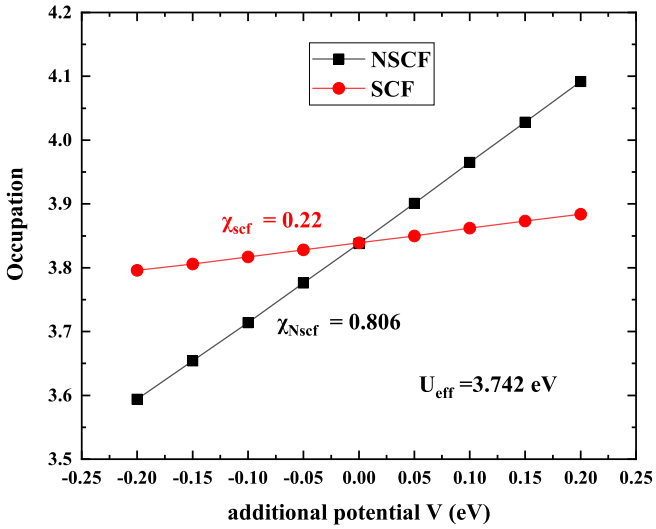


Fig. S14: The calculated effective Hubbard U U_{eff} for the MoTe monolayer using the linear response method.

D. Magnetic ground state of MoTe

To determine the magnetic ground state of the MoTe monolayer, we computed the total energies for the MoTe monolayer for the ferromagnetic (FM) configuration and five representative antiferromagnetic (AFM) configurations, as shown in Fig. S15. Our findings indicate that AFM1 exhibits the lowest energy, suggesting it represents the magnetic ground state.

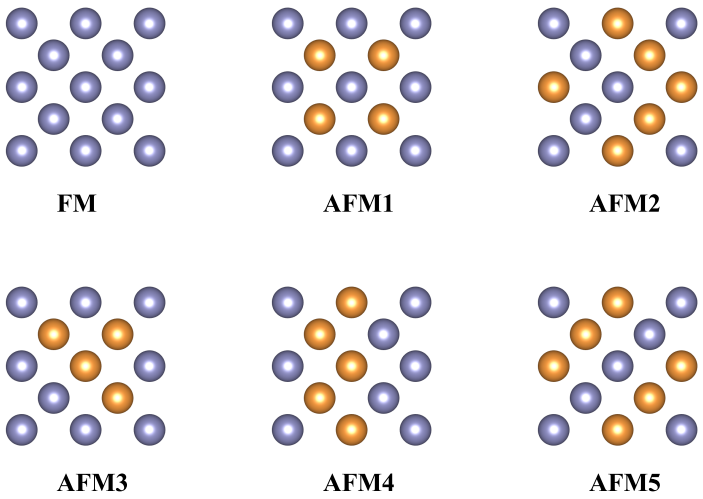


Fig. S15: Top views of the 2 × 2 × 1 supercells of MoTe monolayer with the FM ground state and five typical AFM states. The yellow and blue spheres denote the Mo atoms with opposite magnetization directions. The Te atoms are not depicted here.

Tab. S5: Total energies (eV) for the MoTe monolayer in 2 × 2 × 1 supercells under various magnetic configurations. The total energy of the monolayer with the AFM1 configuration is set to zero.

	FM	AFM1	AFM2	AFM3	AFM4	AFM5
MoTe	0.904	0	1.196	0.974	0.931	1.47

E. Dynamic and thermodynamic stability of MoTe

Figure S16 presents the phonon spectrum of MoTe monolayer, showing no imaginary frequencies throughout the Brillouin Zone, which suggests the structure is dynamically stable. To explore its thermodynamic stability further, ab initio molecular dynamics simulations were performed at 300 K, as illustrated in Fig. S17. The simulations reveal that total energies fluctuate around a constant value, and the monolayer do not change to other structure after simulation, indicating it is thermodynamically stable.

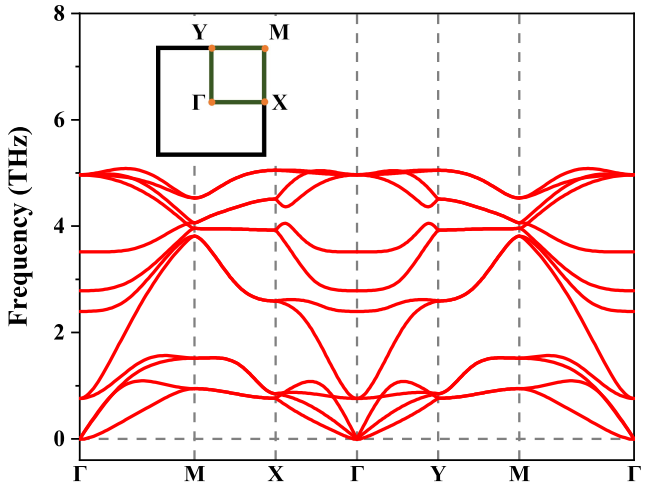


Fig. S16: Phonon spectrum of MoTe monolayer.

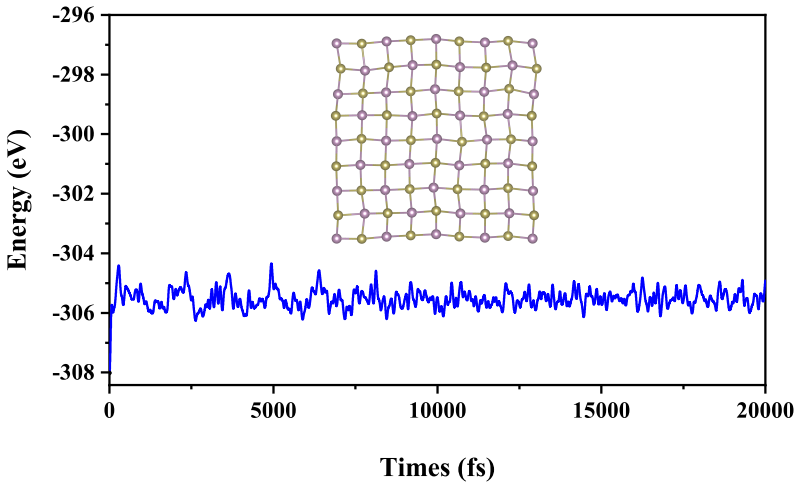


Fig. S17: Fluctuations in the total energy of MoTe during ab-initio molecular dynamics simulations, along with a snapshot captured at the end of the simulations.

F. Energy band structures with spin-orbit coupling for MoTe

In order to investigate the impact of spin-orbit coupling, we additionally compute the energy band structures incorporating spin-orbit coupling, as depicted in Fig. S18. It is evident that spin-orbit coupling minimally affects spin splitting under an external electric field, as observed from the comparison with the energy band structure in Fig. S11.

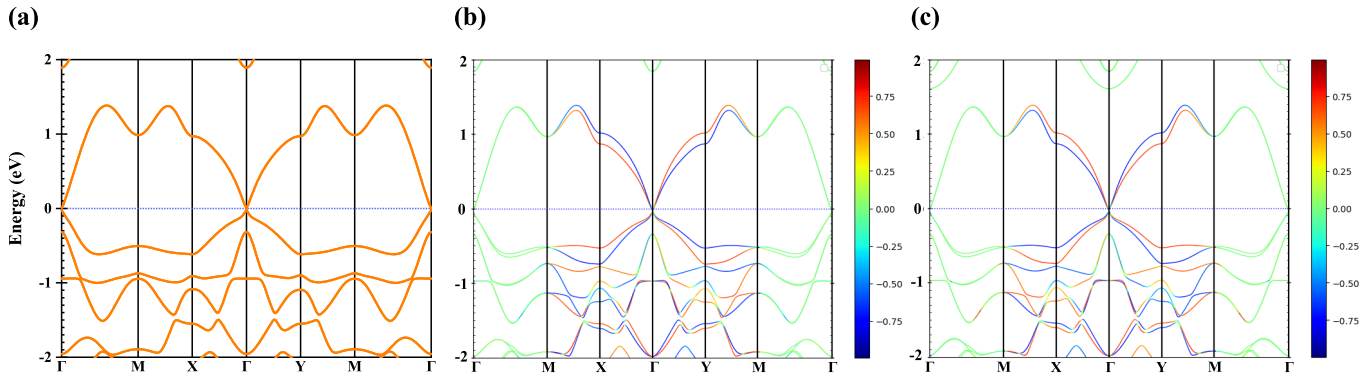


Fig. S18: The energy band structures with spin-orbit coupling are illustrated under three conditions: (a) without an electric field, (b) with an electric field of +0.3 V/Å, and (c) with an electric field of -0.3 V/Å. The spin component in the z-direction is projected for cases of energy band structure with finite electric field.

G. Potential experimental synthesis pathways of MoTe.

The MoTe share the same crystal structure with the experimental synthesized FeSe and CoSe[23] but it has not been synthesized in experiments. In fact, some studies reveal that these two monolayer also host antiferromagnetic ground states[24, 25], so these two materials can also realize the magnetic transition from antiferromagnets to altermagnets. We have calculated the energies of the monolayer with different magnetic orders of Mo atoms, and the results indicate that the antiferromagnetic coupling of nearest neighbor Mo atoms is the magnetic ground state of the 2D monolayer. The first-principles phonon spectrum and molecular dynamics simulations demonstrate that the material is dynamically and thermodynamically stable. These results indicate that it may be synthesized under appropriate experimental conditions.

Next, let's discuss the possible experimental synthesis conditions. It is revealed that the 2D materials growth strongly rely on the interplay between the symmetries of the 2D material and the substrate[26]. It is proposed that, only if the symmetry group of the substrate is a subgroup of that of the 2D material, orientational uniformity of 2D material islands on a substrate can be realized. Take the 2D FeSe and 1T-$FeSe_2$ as an example. Both of these two 2D materials have been synthesized in experiments. The former is a high-temperature superconductor and can be grown on the (001) surface of SrTiO3 and the symmetry of the surface match with the 2D FeSe monolayer with square lattice. For the latter, it is grown on the SiC(0001) substrates with C_3 symmetry, which match with the crystal structure of 1T-$FeSe_2$ monolayer. Therefore, because the 2D MoTe host C_{4v} symmetry, we must choose a square surface to grow it. In addition, the matching of lattice constants is also crucial. Given that the in-plane crystal constant of MoTe is 4.44 Å, the crystal constants of 2×2 $SrTiO_3$ with the 001 orientation align well with these values, with only a 0.5% mismatch.

H. Limitation of applying vertical electric field

The method of applying external electric field to induce the magnetic transition is limited to 2D systems, which is mainly due to the following two reasons:

1. Material Dimensionality and Field Penetration

In 2D systems, the electric field can uniformly influence the entire material layer owing to its atomic-scale thickness, allowing effective modulation of electronic and magnetic states. In contrast, 3D materials possess thicker volumes and more complex internal structures, which can lead to non-uniform electric field penetration and diminished control over the entire bulk. Moreover, the electric field applied to the periodic direction can break the periodic of crystal and corresponding Bloch electronic states.

2. Screening Effects

In 3D materials, free carriers and polarization effects can screen external electric fields, reducing their effectiveness in modulating magnetic phases. 2D systems, with reduced screening due to their limited dimensionality, respond more readily to external electric stimuli.

In addition to the factors mentioned earlier that confine the external electric field to affect only two-dimensional materials, the modulation of antiferromagnetic 2D materials by an external electric field is also limited by the requirement that their opposing spin sublattices must reside in the same plane perpendicular to the electric field. This is because the electric potential corresponding to the field varies across different planes. If the opposing spin sublattices are positioned in different planes, the varying potentials disrupt any symmetry operations that would connect the opposing spin sublattices.

I. Energy band structures of other 2D monolayers with vertical electric field

The energy band structures of analogous monolayers such as CoSe, CrS, FeAs, OsTe, and VSb are illustrated in Fig. S19. It is shown that an external vertical electric field in opposite directions can also induce opposite spin splitting in these monolayers.

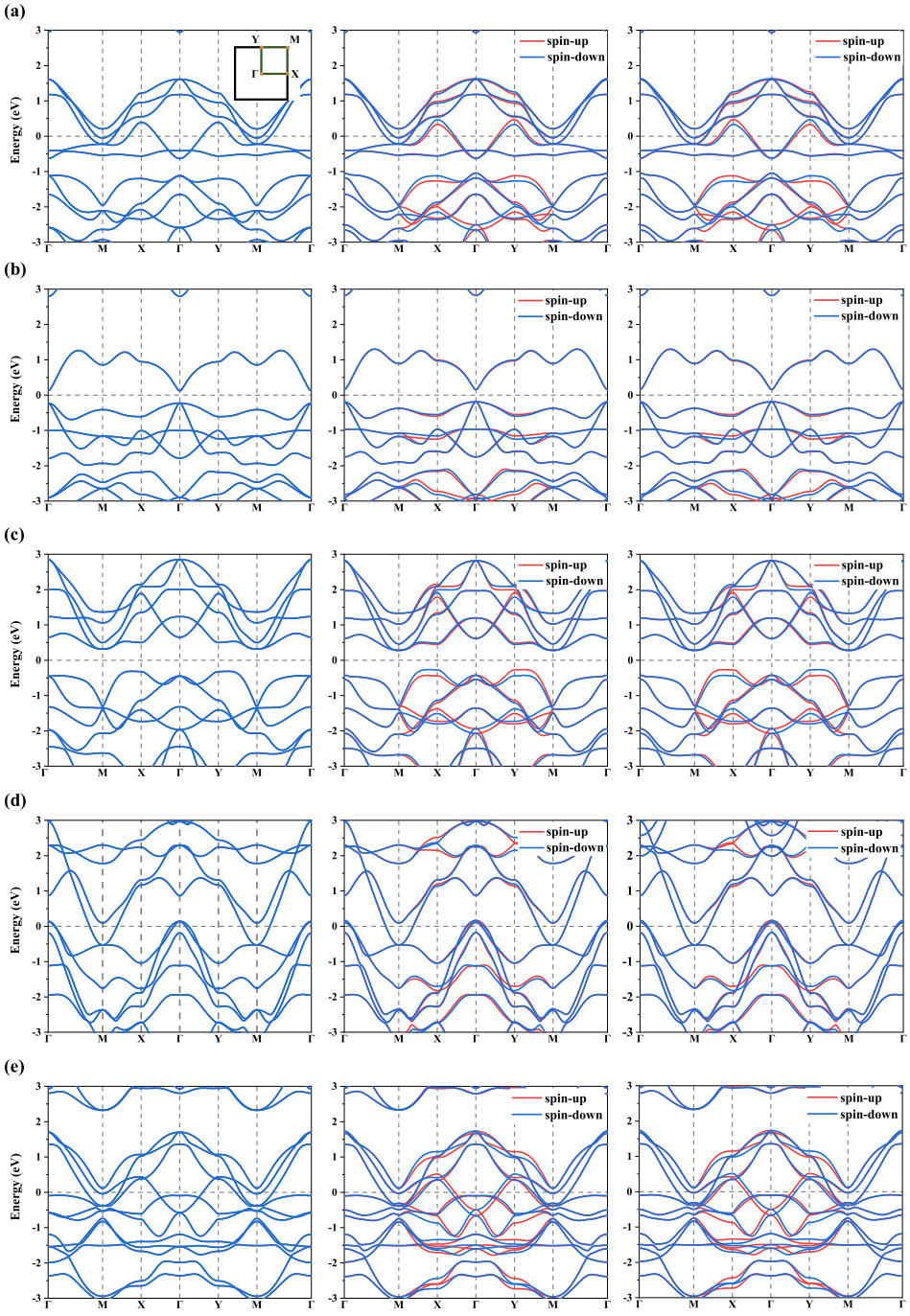


Fig. S19: Energy band structures of 2D antiferromagnetic materials (a) CoSe, (b) CrS, (c) FeAs, (d) OsTe, and (e) VSb are presented under three conditions: without an electric field, with an electric field of 0.1 V/Å, and with an electric field of -0.1 V/Å.

III. The practical feasible approach from antiferromagnets to altermagnets

Here we present several feasible methods for the magnetic transition from antiferromagnets to altermagnets, including Floquet engineering by applying light fields, and surface engineering by cutting surface along specific surface orientation.

A. Floquet engineering

While external electric fields can break the PT symmetry in antiferromagnetic systems, the electronic lattice potential experienced by opposite-spin sublattices may remain equivalent, potentially preserving spin degeneracy in the energy bands. To overcome this limitation, a more versatile approach, namely Floquet engineering, can be employed to manipulate system symmetries through periodic driving. In recent years, Floquet engineering, which utilizes time-periodic light fields to realize novel quantum states, has garnered significant attention. At its core, this method leverages light-matter interactions to dynamically alter the symmetry properties of a system, enabling access to otherwise inaccessible quantum phases. Here, we propose that a magnetic transition from antiferromagnetism to altermagnetism can be induced via tailored light-matter coupling in antiferromagnetic systems. Specifically, applying circularly polarized light to a PT-symmetric antiferromagnet can effectively break the PT symmetry, thereby enabling the emergence of an altermagnetic phase.

1. Introduction of Floquet engineering theory

To reveal the light-induced magnetic transition in antiferromagnetic materials, we constructed the tight-binding (TB) Hamiltonian as

$$\mathcal{H}(\mathbf{R}) = \sum_{m,n}\sum_{i,j} t_{ij}^{mn} C_i^{\dagger}(\mathbf{R}_m) C_j(\mathbf{R}_n) + H.c., \tag{4}$$

where $C_i^{\dagger}(\mathbf{R}_m)(C_j(\mathbf{R}_n))$ is the creation (annihilation) operators at orbital i (j) on site $\mathbf{R}_m$ ($\mathbf{R}_n$), t_{ij}^{mn} are the hopping integrals between orbital i at site $\mathbf{R}_m$ and orbital j at site $\mathbf{R}_n$. When a time-periodic and space-homogeneous monochromatic light field $\mathbf{A}(t)$ is applied to materials, the time-dependent hopping integrals are obtained by using the Peierls substitution $\mathbf{k} \to \mathbf{k} + \mathbf{A}(t)$. Therefore, the time-dependent hopping integrals are written as

$$t_{ij}^{mn}(t) = t_{ij}^{mn} \exp(i\frac{e}{\hbar}\mathbf{A}(t)\cdot\mathbf{d}_{ij}), \tag{5}$$

where $\mathbf{d}_{ij}$ is the relative position vector between orbital i at site $\mathbf{R}_m$ and orbital j at site $\mathbf{R}_n$. Thus we obtained the time-dependent Hamiltonian as

$$\mathcal{H}(\mathbf{k},t) = \sum_{m,n}\sum_{i,j} t_{ij}^{mn}(t)\exp(i\mathbf{k}\cdot\mathbf{R}_{m-n}) C_i^{\dagger}(\mathbf{k},t) C_j(\mathbf{k},t) + H.c. \tag{6}$$

Due to the periodic translation symmetry in the time and lattice, Floquet theorem can be effectively applied to the time-dependent Hamiltonian. Thus we express the Floquet-Bloch creation (annihilation) operator as

$$C_i^{\dagger}(\mathbf{k},t) = \sum_{m,n}\sum_{a=-\infty}^{\infty} C_{ai}^{\dagger}(\mathbf{R}_m)\exp(i\mathbf{k}\cdot\mathbf{R}_m - ia\Omega t) \tag{7}$$

$$C_j^{\dagger}(\mathbf{k},t) = \sum_{m,n}\sum_{b=-\infty}^{\infty} C_{bj}(\mathbf{R}_n)\exp(-i\mathbf{k}\cdot\mathbf{R}_n + ib\Omega t) \tag{8}$$

Thus, we gain an effective static Hamiltonian in frequency and momentum space

$$\mathcal{H}^f(\mathbf{k},\Omega) = \sum_{i,j}\sum_{a,b}[\mathcal{H}_{ij}^{a-b}(\mathbf{k},\Omega) + (a-b)\hbar\Omega\delta_{ij}\delta_{ab}] C_{ai}^{\dagger}(\mathbf{k}) C_{bj}(\mathbf{k}) + H.c., \tag{9}$$

where $\hbar\Omega$ represents the photonic energy, Ω represents the photonic frequency, and the matrix $\mathcal{H}_{ij}^{a-b}(\mathbf{k},\Omega)$ is written as

$$\mathcal{H}_{ij}^{a-b}(\mathbf{k},\Omega)=\sum_{i,j}t_{ij}^{mn}\exp(i\mathbf{k}\cdot\mathbf{R}_{m-n})\left(\frac{1}{T}\int_0^T\exp(i\frac{e}{\hbar}\mathbf{A}(t)\cdot\mathbf{d}_{ij})\exp(i(a-b)\Omega t)dt\right). \tag{10}$$

Where T ($T=2\pi/\Omega$) represent the time period, the (a, b) is the Floquet index ranging from $-\infty$ to $+\infty$. Then the Floquet-Bloch Hamiltonian can be described in following form

$$\mathcal{H}^f(\mathbf{k},\Omega)=\begin{pmatrix}\ddots & \vdots & \vdots & \vdots & \iddots\\ \cdots & \mathcal{H}_0-\hbar\Omega & \mathcal{H}_{-1} & \mathcal{H}_{-2} & \cdots\\ \cdots & \mathcal{H}_1 & \mathcal{H}_0 & \mathcal{H}_{-1} & \cdots\\ \cdots & \mathcal{H}_2 & \mathcal{H}_1 & \mathcal{H}_0+\hbar\Omega & \cdots\\ \iddots & \vdots & \vdots & \vdots & \ddots\end{pmatrix}. \tag{11}$$

To calculate the matrix $\mathcal{H}_{ij}^{a-b}(\mathbf{k},\Omega)$ in Eq. (S7), we give $\mathbf{A}(t)$ and $\mathbf{d}_{ij}$ as the following general forms

$$\mathbf{A}(t)=[A_x sin(\Omega t+\Phi_1),A_y sin(\Omega t+\Phi_2),A_z sin(\Omega t+\Phi_3)], \tag{12}$$

$$\mathbf{d}_{ij}=[d_x,d_y,d_z], \tag{13}$$

where A_x, A_y, and A_z (d_x, d_y, and d_z) represent the amplitude of potential vector (the component of the position vector) along the x, y, and z direction, respectively. The phases Φ_1, Φ_2, and Φ_3 are the related initial phases of the component of the potential vector. Then, we can re-write the matrix element $\mathcal{H}_{ij}^{q}(\mathbf{k},\Omega)$ ($q=a-b$) as

$$\mathcal{H}_{ij}^{q}(\mathbf{k},\Omega)=\sum_{i,j}t_{ij}^{mn}\exp(i\mathbf{k}\cdot\mathbf{R}_{m-n})\exp(-iq\Phi)\cdot i^q J_q(\frac{e}{\hbar}A_{max}), \tag{14}$$

where J_q is the q-th Bessel function

$$J_q(\frac{e}{\hbar}A_{max})=\exp(iq\Phi)\left(\frac{1}{T}\int_0^T\exp(i\frac{e}{\hbar}A_{max}\sin(\Omega t+\Phi)\exp(iq\Omega t)dt\right). \tag{15}$$

A_{max} represents the amplitude of the light-field potential vector, Φ represents the angle between the position vector and the x-axis, and A_{max} and Φ can be calculated using the following formula

$$\begin{aligned}&A_{max}=\sqrt{\left(A_xd_x\sin\Phi_1+A_yd_y\sin\Phi_2+A_zd_z\sin\Phi_3\right)^2+\left(A_xd_x\cos\Phi_1+A_yd_y\cos\Phi_2+A_zd_z\cos\Phi_3\right)^2},\\ &\Phi=\arctan\left(\frac{A_xd_x\sin\Phi_1+A_yd_y\sin\Phi_2+A_zd_z\sin\Phi_3}{A_xd_x\cos\Phi_1+A_yd_y\cos\Phi_2+A_zd_z\cos\Phi_3}\right).\end{aligned} \tag{16}$$

2. Results of antiferromagnetic honeycomb lattice under Floquet engineering

When the left circularly polarized light is replaced with right circularly polarized light, the spin splitting can be reversed, as illustrated in Figs. S20(a) and S20(b). Additionally, the magnitude of spin splitting can be adjusted by altering the light intensity. For example, the spin splitting at the K point can increase from 0.406 eV to 0.454 eV when the light intensity changes from 1.1/Å to 1.2/Å, as shown in Figs. S20(c) and S20(d).

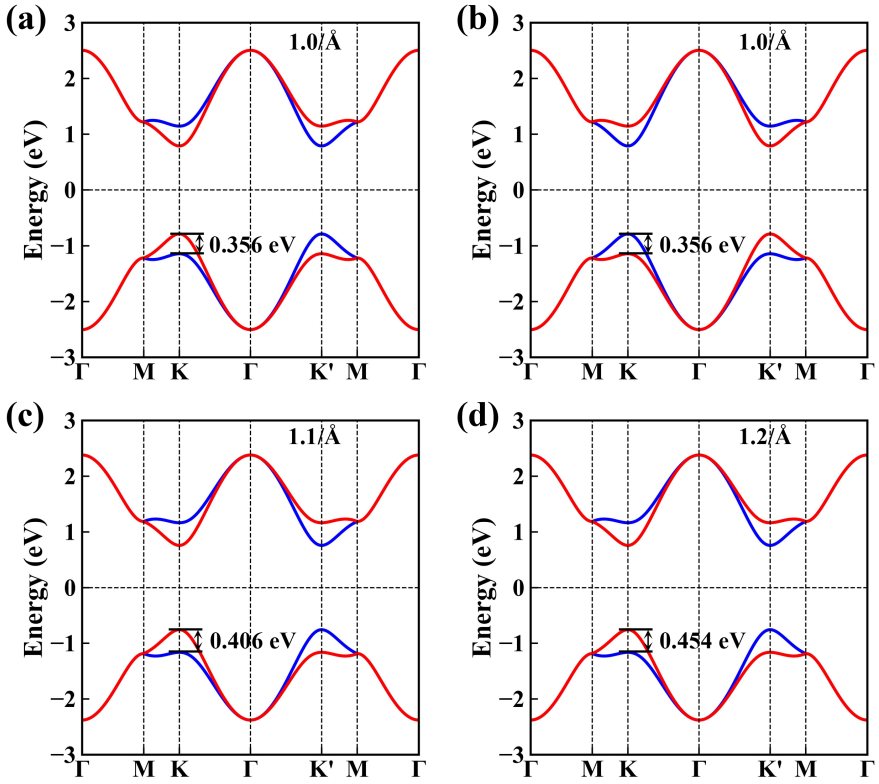


Fig. S20: The energy band structure of the antiferromagnetic honeycomb lattice under (a) left circularly polarized light and (b) right circularly polarized light with an intensity of 1.0/Å. The energy band structure of the antiferromagnetic honeycomb lattice under left circularly polarized light with intensities of (c) 1.1/Å and (d) 1.2/Å.

3. *Honeycomb antiferromagnet MnSr via light-field driven Floquet engineering*

In the main text, we have demonstrated the effectiveness of periodically driven Floquet engineering using both a theoretical honeycomb antiferromagnetic model and the three-dimensional (3D) antiferromagnetic material Cr_2O_3. Here, we further investigate the influence of light fields on the symmetry and magnetic state of two-dimensional (2D) antiferromagnetic systems. As illustrated in Figs. S21(a) and S21(b), MnSr is a representative 2D honeycomb antiferromagnet in which the magnetic atoms form a planar honeycomb lattice. The opposite-spin sublattices in MnSr share identical electronic lattice potentials, and the system is described by the spin point group ${}^{2}6/{}^{1}m^{2}m^{1}m$, characteristic of a PT-symmetric antiferromagnet. Consequently, its band structure exhibits spin degeneracy, as shown in Fig. S21(c) (Since we are only concerned with the spin splitting under the action of light field, we only consider the d orbitals of Mn atoms). MnSr hosts a Néel-type antiferromagnetic ground state that is dynamically stable, as evidenced by the phonon spectrum without imaginary frequencies in Fig. S21(d). For this class of PT-symmetric antiferromagnets, conventional electric field methods are ineffective in lifting the spin degeneracy due to the preservation of identical lattice potentials across spin sublattices. However, upon the application of circularly polarized light, PT symmetry is broken, and the light-induced anisotropy introduces distinct electronic lattice potentials for the two spin sublattices. Specifically, when left circularly polarized light with a photon energy of $\hbar\Omega = 10$ eV and intensity of 0.35/Å is incident along the z-direction, the Mn atoms with opposite spins experience opposite potentials, resulting in altermagnetic spin splitting [Figs. S21(f) and S21(g)]. Under the influence of the light field, the system transitions from a static magnetic state characterized by the spin point group ${}^{2}6/{}^{1}m^{2}m^{1}m$ to a nonequilibrium Floquet state with spin point group ${}^{1}\bar{6}^{2}m^{2}2$. As depicted in Fig. S21(e), this transition implies that the spin-up and spin-down sublattices become connected via three vertical mirror symmetries.

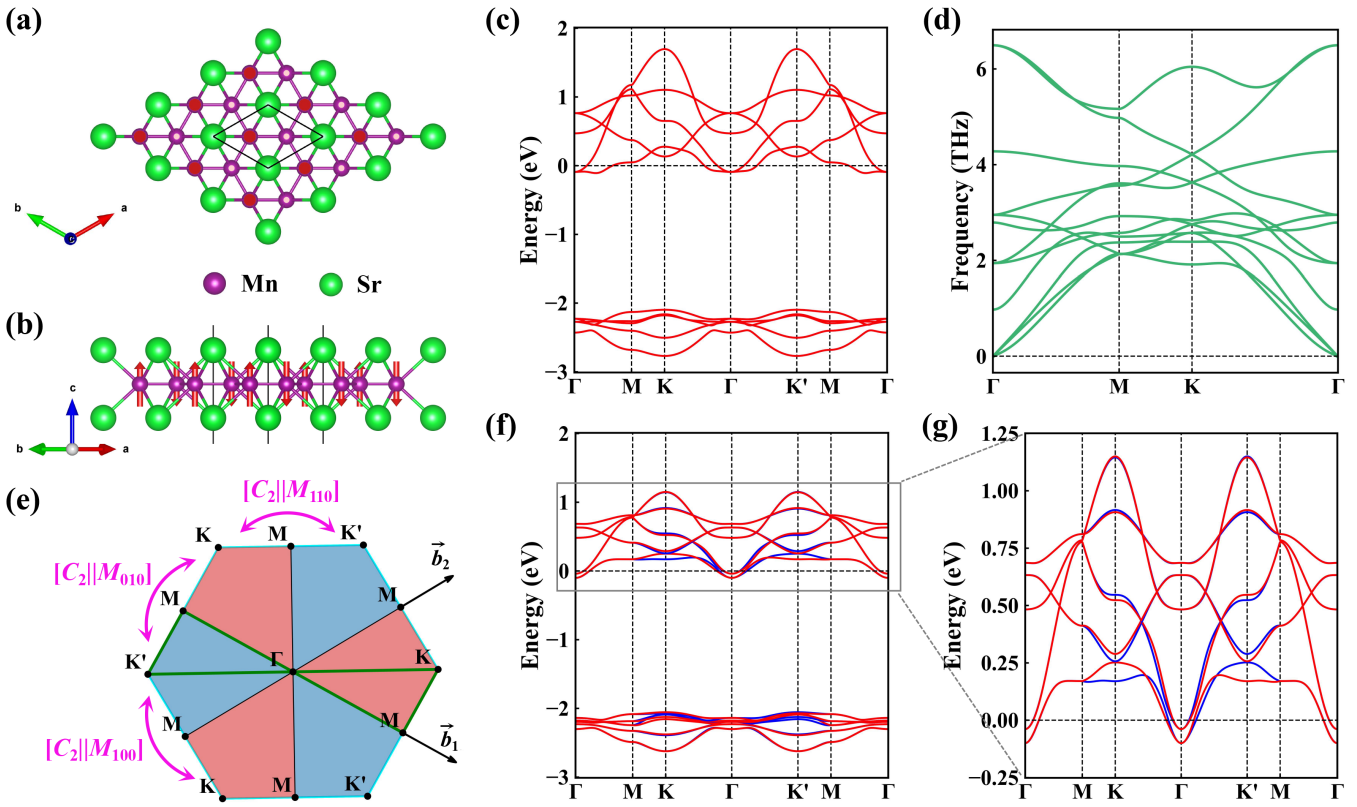


Fig. S21: Honeycomb antiferromagnet MnSr. (a,b) Crystal structure of antiferromagnet MnSr. (c), (d), and (e) The band structure, phonon spectrum, and Brillouin zone, respectively. (f) and (g) The band structure under left circularly polarized light with the light intensity 0.35/Å.

4. Brillouin zone and bulk bands for Cr_2O_3 and KMnP

Here, we presents the three-dimensional Brillouin zones and bulk electronic band structures of Cr_2O_3 and KMnP in Fig. S22. Both antiferromagnetic materials exhibit semiconducting behavior, with calculated band gaps of 1.36 eV for Cr_2O_3 and 1.34 eV for KMnP.

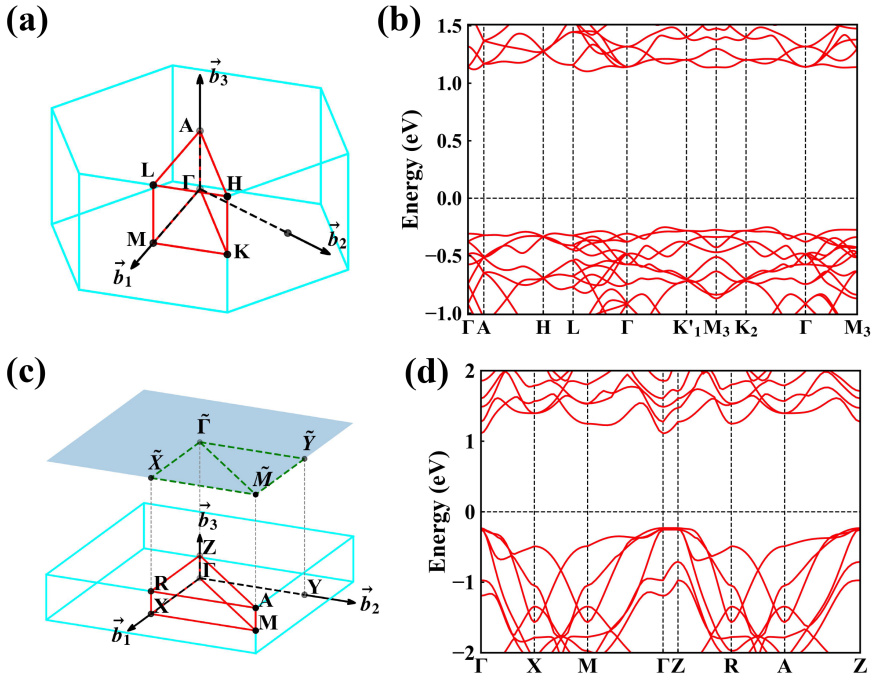


Fig. S22: The Brillouin zone and bulk band structure of (a,b) Cr_2O_3 and (c,d) KMnP.

B. Surface engineering

Here, we consider materials composed primarily of light elements and neglect relativistic spin-orbit coupling (SOC) as an approximation. As illustrated in Fig. S23(a), a bulk antiferromagnetic material typically contains two spin sublattices, whose symmetries are described by a spin space group (SSG) $\mathbf{G} = [\mathbf{R}_i \| (\mathbf{R}_j | \boldsymbol{t})]$. Upon cleaving the material along a specific Miller plane (hkl), the surface interfaces with vacuum. We define $\hat{\mathbf{n}}$ as the unit vector normal to the surface, which plays a key role in characterizing the surface symmetry. According to S. F. Weber et al's work[27], the symmetry group of a surface, denoted as $\mathbf{G}_{\hat{\mathbf{n}}} = [\mathbf{R}_i \| (\mathbf{R}_j^{\hat{\mathbf{n}}} | \boldsymbol{t}^{\perp \hat{\mathbf{n}}})]$, is a subgroup of the bulk symmetry group, as shown in Fig. S23(b). For simplicity, we focus here on collinear antiferromagnets and altermagnets, though the general approach discussed can be extended to noncollinear systems. Because we concentrate on surfaces where opposite spin sublattices coexist and are magnetically compensated, i.e., the net magnetization vanishes. Hence, these surfaces do not exhibit equilibrium magnetization [27, 28]. Since altermagnetic spin splitting is primarily governed by the point group operations, we restrict our discussion to the spin point group of the surface, denoted by $\mathbf{g}_{\hat{\mathbf{n}}} = [\mathbf{R}_i \| \mathbf{R}_j^{\hat{\mathbf{n}}}]$. If the surface spin point group permits altermagnetism, meaning it lacks symmetry operations such as $\boldsymbol{t}T$ or PT that would otherwise relate opposite spin sublattices, then surface altermagnetism can emerge. Furthermore, surface altermagnetism can also be achieved through surface engineering by constructing a 2D counterpart from a specific crystal face of the 3D material, as illustrated in Fig. S23(c). In this configuration, both the top and bottom surfaces of the extracted 2D layer are exposed to vacuum, thereby breaking $\boldsymbol{t}T$ symmetry while retaining inversion symmetry.

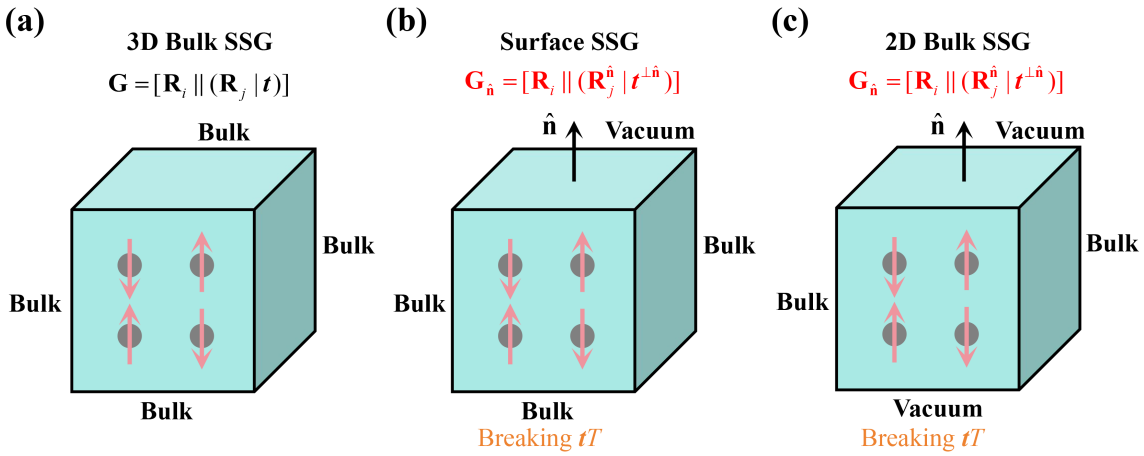


Fig. S23: (a) Spin space group description of 3D collinear antiferromagnetic materials. (b) Surface spin space group descriptions after cutting a semi-infinite surface. (b) and (c) Spin space group descriptions after cutting a two-dimensional slab.

1. *Surface engineering applied to $\boldsymbol{tT}$-PT-symmetric antiferromagnetic $CaFeO_2$*

Here, we demonstrate the emergence of altermagnetism in the purely $\boldsymbol{tT}$-symmetric antiferromagnetic material $CaFeO_2$ through the introduction of a surface cut along a specific crystallographic orientation.

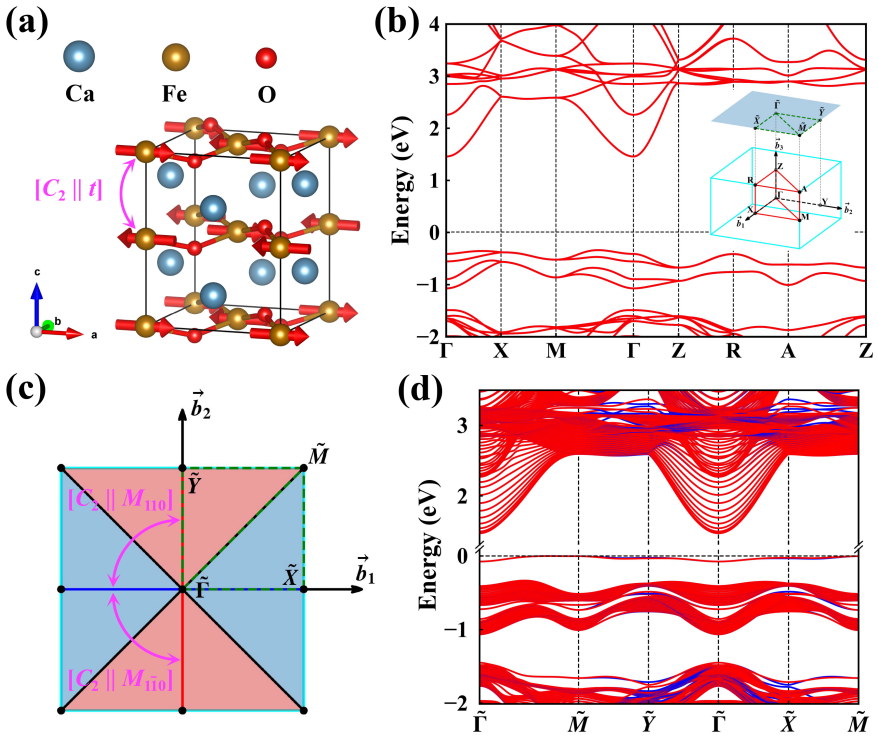


Fig. S24: Antiferromagnetism to altermagnetism by Surface engineering in $\boldsymbol{tT}$ antiferromagnetic materials $CaFeO_2$. (a) The crystal structure and (b) bulk band of the antiferromagnetic materials $CaFeO_2$, respectively. Inset shows the 3D Brillouin zone. (c) The 2D BZ plane reveals altermagnetic spin splitting. (d) Slab band structure of $CaFeO_2$ with 15 layers along the z-direction, where blue and red represent spin-up and spin-down states, respectively.

C. Combination of surface engineering and Floquet engineering

While Floquet engineering provides a versatile approach to manipulating system symmetries, capable of breaking rotational, mirror, time-reversal, and spatial inversion symmetries via tailored light fields, it remains ineffective against the $\boldsymbol{tT}$ symmetry inherent in certain antiferromagnetic systems. As a result, Floquet engineering alone cannot drive the magnetic transition from a $\boldsymbol{tT}$-symmetric antiferromagnet to an altermagnet. In contrast, a combined approach that integrates surface engineering with Floquet engineering is capable of simultaneously breaking both $\boldsymbol{tT}$ and PT symmetries. This dual-symmetry-breaking strategy enables the transition from antiferromagnetism to altermagnetism, providing a general and robust method for realizing altermagnetism in $\boldsymbol{tT}$-symmetric antiferromagnetic systems.

1. *The combination method applied to $\boldsymbol{tT}$-symmetric antiferromagnetic NiO*

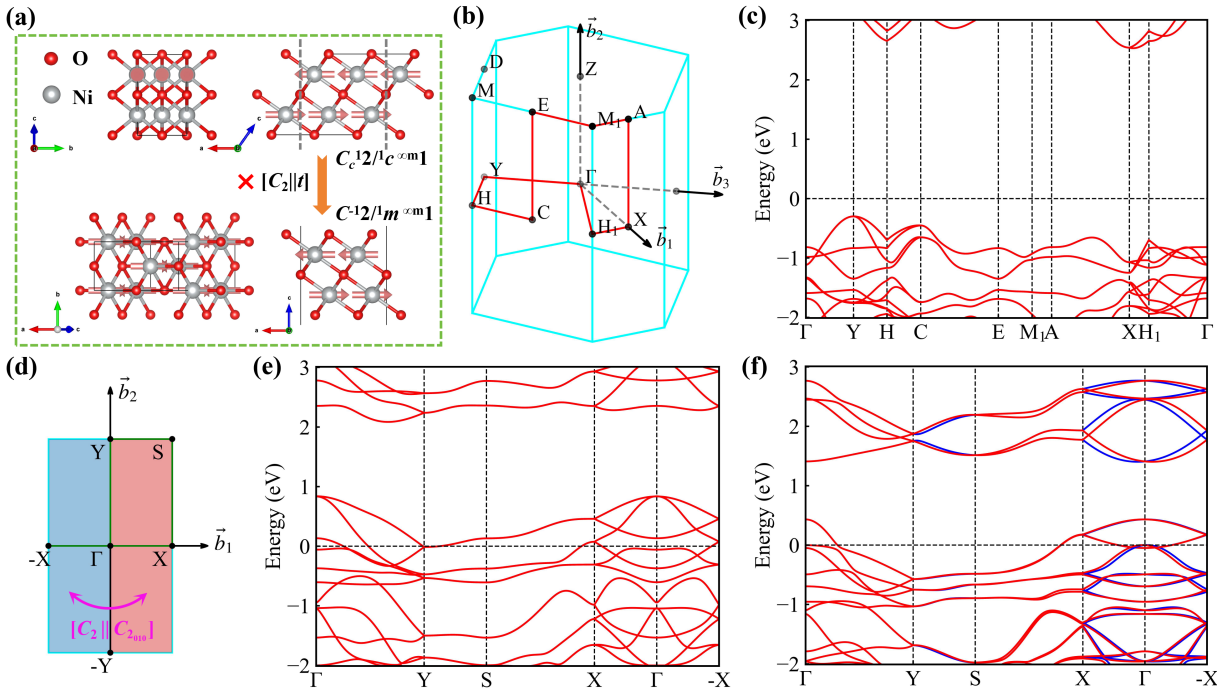


Fig. S25: Antiferromagnetism to altermagnetism by the combination of Surface engineering and Floquet engineering in typical $\boldsymbol{tT}$ antiferromagnet NiO. (a) The crystal structures of the magnetic primitive cell of antiferromagnet NiO, and the construction process of the specific surface 2D antiferromagnetic counterpart. (b) The Brillouin zones and (c) the band structure of 3D NiO, respectively. (d) 2D Brillouin zone with altermagnetic spin splitting. The energy band structure of 2D antiferromagnet NiO with left circularly polarized light intensity (e) 0.0 and (f) 0.5/Å.

To explicitly demonstrate the magnetic transition from a $\boldsymbol{tT}$-symmetric antiferromagnet to an altermagnet, we take the widely studied cubic antiferromagnet NiO[29] as a representative example. We adopt the experimentally determined rocksalt crystal structure[30] and apply an effective Hubbard U of 4.6 eV to the Ni $3d$ orbitals within the GGA + U framework. In its paramagnetic phase, NiO crystallizes in the cubic space group $Fm\bar{3}m$. Upon entering the antiferromagnetic phase, the Ni local magnetic moments align along the $[11\bar{2}]$ direction, reducing the symmetry to a monoclinic magnetic space group C_c2/c (spin space group $C_c^12/^1c^{\infty m}1$) for the magnetic primitive cell, as illustrated in Fig. S25(a), with its Brillouin zone and high-symmetry k-paths shown in Fig. S25(b). Due to the presence of $\boldsymbol{tT}$ symmetry in bulk NiO, its energy bands are fully spin-degenerate throughout the 3D Brillouin zone, as shown in Fig. S25(c). To break this $\boldsymbol{tT}$ symmetry, we employ surface engineering to construct a 2D antiferromagnetic NiO slab. This reduces the symmetry from the spin space group $C_c^12/^1c^{\infty m}1$ to the $C^{-1}2/^1m^{\infty m}1$, as depicted in Fig. S25(a). The corresponding spin point group becomes ${}^22/^1m$, which still preserves PT symmetry, resulting in spin-degenerate energy bands, as shown in Fig. S25(e). To further break the PT symmetry and induce altermagnetism,

we apply left-handed circularly polarized light along the b axis. Under the influence of the light field, the symmetry is lowered to the spin point group $^{2}2$, and the electronic bands exhibit clear altermagnetic spin splitting, as shown in Figs. S25(d) and S25(f). Symmetry analysis reveals that the spin-up and spin-down sublattices are connected via the operation $[C_2\|C_{2_{010}}]$, leading to spin splitting that is symmetric with respect to the symmetry $C_{2_{010}}$ in reciprocal space.

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